THE EVEN ICOSAHEDRAL REPRESENTATION OF CONDUCTOR 1951: A CONDITIONAL REDUCTION AND EXPLICIT RESIDUAL BRIDGES

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ABSTRACT. Let ρ be an even icosahedral Artin representation of conductor 1951, the smallest prime conductor of an even icosahedral representation. To our knowledge, Artin's conjecture is not known for any even icosahedral representation of $G_{\mathbb{Q}}$. We record five results. (A) A conditional reduction modelled on Tunnell's proof in the octahedral case: if ρ becomes automorphic over a single quadratic field (for instance an imaginary one), and base change for GL(2) from $\mathbb{Q}$ to the non-normal quintic field L_5 cut out by ρ exists for one specific cuspidal representation, then ρ is modular and $L(\rho, s)$ is entire. A second form replaces the base change hypothesis by automorphy over infinitely many imaginary quadratic fields; conversely, modularity of ρ implies automorphy over all of them. (B) The reduction of ρ modulo $\sqrt{5}$ arises, up to a character twist, from an elliptic curve over a number field of degree at most 4 only if that field is totally imaginary and contains $\sqrt{5}$; in particular never over a quadratic field. (C) Explicit elliptic curves realising this twist over $\mathbb{Q}(i, \sqrt{5})$ and over $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$, obtained from Klein's icosahedral resolvent, certified by exact computation, and modular by a theorem of Caraiani and Newton. (D) The reduction of ρ modulo 2 occurs in $S_2(\Gamma_0(1951), \chi)$ through both of its 2-stabilisations, each with multiplicity one, while for parity reasons no holomorphic eigenform is congruent to ρ modulo 4. It comes from a weight-one Katz form modulo 2 which, because ρ is even, is not congruent to any weight-one form in characteristic 0. (E) The same constructions work for all six even icosahedral representations of prime conductor below 10^4, which Doud and Moore listed. The trace-zero lattice of the quintic field is always the root lattice A_4 scaled by the conductor, so the principal quintic that starts Klein's construction exists over every imaginary quadratic field. We certify the analogues of (C) for all six representations and of (D) for three of them, and for their adjoint representations we check Stark's conjecture numerically in rank 3. We also record a certified positivity of Weil's quadratic form for $L(\rho, s)$ on finite-dimensional spaces of test functions, computed from the primes alone, and numerical data on the p-adic L-functions of ρ, for which the p-adic analogue of Artin's conjecture holds by work of Greenberg and Wiles. We compare the base change hypothesis in (A) with the known cubic and quartic cases, including recent work of Wang. The converse-theorem route to it would need automorphic induction from L_5 (for the trivial character, automorphy of the four-dimensional representation $\rho_0 \otimes \rho_0''$ of A_5), so that route is entangled with icosahedral automorphy. Nothing here proves that $L(\rho, s)$ is entire unconditionally.

1. INTRODUCTION

Let $f_5 = x^5 - x^4 - 780x^3 - 1795x^2 + 3106x + 344$, let $L_5 = \mathbb{Q}[x]/(f_5)$, and let M be the Galois closure of L_5, so that $\mathrm{Gal}(M/\mathbb{Q}) \cong A_5$ and M is totally real. Doud and Moore [17] showed that the smallest prime conductor of an even icosahedral Galois representation is 1951. We write $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{C})$ for the representation of conductor 1951 with projective kernel field M and determinant the character χ of §2, which has order 5 and conductor 1951. Then $\rho = \rho_0 \otimes \chi^3$, where $\rho_0\colon G_{\mathbb{Q}} \to \mathrm{SL}_2(\mathbb{Z}[\zeta_5])$ has image isomorphic to $\mathrm{SL}(2, 5)$ and traces in $\mathbb{Z}[\varphi]$, $\varphi = (1 + \sqrt{5})/2$. (It is not realisable over $\mathbb{Q}(\sqrt{5})$, since finite subgroups of $\mathrm{SL}_2(\mathbb{R})$ are cyclic.) Complex conjugation acts on ρ by a scalar, so ρ is even.

Artin's conjecture is known for two-dimensional representations with solvable projective image [44, 59], and for odd two-dimensional representations of $G_{\mathbb{Q}}$ as a consequence of Serre's conjecture [39]. For even icosahedral representations of $G_{\mathbb{Q}}$ no case is known [7]. By Booker [3], $L(\rho, s)$ is

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entire if and only if ρ is modular, i.e. corresponds to a cuspidal automorphic representation of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$; this would be a Maass form of eigenvalue $1/4$, level 1951 and nebentypus χ. There is strong numerical evidence for such a form. Hejhal's method [30] on $\Gamma_0(1951)$, with input only the level, the nebentypus, the parity and the eigenvalue $1/4$, produces a form whose Fourier coefficients agree with the Dirichlet coefficients of $L(\rho,s)$ to within $3\cdot 10^{-8}$ for $n\le 800$ and $6\cdot 10^{-7}$ for $n\le 2000$. Independently of Hejhal's method, the odd series $\sqrt{y}\sum_n a_n K_0(2\pi n y)\sin(2\pi n x)$, built from the Dirichlet coefficients a_n of $L(\rho,s)$ alone, is invariant under $\Gamma_0(1951)$ with nebentypus χ to within $1.7\cdot 10^{-10}$ at 24 random elements. The series of the $\sqrt{5}$-conjugate $\rho''=\rho_0''\otimes\chi$ (Remark 6.3) is invariant with nebentypus χ^2 to within $1.1\cdot 10^{-10}$. Replacing K_0 by K_{ir}, the values of r^2 that would cancel the defects to first order are below $4\cdot 10^{-11}$, so the series singles out the eigenvalue $1/4$. These computations are evidence only and are not used below. Booker [5] shows that a Galois-conjugate pair of non-dihedral cusp forms on $\mathrm{GL}(2)$ with Hecke eigenvalues in $\mathbb{Z}[\varphi]$ has several properties predicted for icosahedral representations, without producing a Galois representation.

The purpose of this note is to record what can presently be proved about ρ from the Langlands side, and to state precisely two inputs that would suffice for modularity.

- Theorem A (§3) is a conditional reduction: automorphy of ρ over one quadratic field together with one instance of non-normal quintic base change implies modularity. It is Tunnell's argument [59] moved up one degree, with the central character of each quadratic descent fixed by an adjoint pole argument. Theorem A′ replaces the base change by automorphy over infinitely many imaginary quadratic fields, and is then an equivalence. Theorem A″ needs automorphy over only one quadratic and one cyclic cubic field. It uses a totally real field, over which ρ is again even, and Proposition 3.9 shows that arguments through local components cannot replace it by finitely many fields without real places.
- Theorem B (§4) determines over which fields of degree at most 4 the reduction of ρ modulo $\sqrt{5}$ can come from an elliptic curve.
- Theorem C (§5) gives two explicit modular elliptic curves carrying it.
- Theorem D (§6) describes the reduction modulo 2 inside $S_2(\Gamma_0(1951),\chi)$, and shows that it comes from a weight-one Katz form modulo 2 that does not lift to characteristic 0.
- Section 8 treats the six representations of prime conductor below 10^4. Lemma 8.1 identifies the trace-zero lattice, Theorems E and F give the analogues of Theorems C and D, and Remark 8.4 checks Stark's conjecture numerically in rank 3.
- Section 7 records two later computations: positivity of Weil's quadratic form for $L(\rho,s)$ on finite-dimensional spaces, certified in ball arithmetic (Proposition 7.1), and special values and p-adic L-functions of ρ (Lemma 7.3, Remark 7.4).

The hypotheses of Theorems A, A′ and A″ are open, and nothing in this note proves that $L(\rho,s)$ is entire. Section 9 lists questions on which expert input would be useful. All computations are exact unless labelled numerical; scripts and data are in the supplement.

2. Notation

- $\chi\colon(\mathbb{Z}/1951)^\times\to\mu_5$ is the character with $\chi(3)=\zeta_5^2$, where 3 is a primitive root modulo 1951 (Conrey label 1951.908). It is even, and we also view it as a Galois character.
- $t_p=\operatorname{tr}\rho_0(\mathrm{Frob}_p)\in\{0,\pm1,\pm2,\pm\varphi,\pm(\varphi-1)\}$ and $a_p=\operatorname{tr}\rho(\mathrm{Frob}_p)=\chi(p)^3t_p$ for $p\neq 1951$. The t_p were computed for $p<3.2\cdot 10^6$ from Frobenius classes in the field of ρ_0. We used the following inputs.
  - The splitting of p in L_5 and in the field K_{24} of degree 24 fixed by a subgroup of order 5 of $\mathrm{Gal}(\tilde{M}/\mathbb{Q})\cong\mathrm{SL}(2,5)$, where $\tilde{M}$ is the field of ρ_0. At the primes dividing the index of the defining polynomial (e.g. $2,7,71,137$ for f_5) we used the maximal order.
  - Resolvents separating the two classes of elements of order 5.

  - A local computation at 1951 that distinguishes ρ_0 from its $\sqrt{5}$-conjugate ρ_0''.

  The two choices involved are certified exactly. First, K_{24} is a cyclic quartic extension of K_6 (below) with discriminant $1951^{16} = \mathrm{disc}(K_6)^4$, so it is unramified over K_6 at all finite places; this fixes ρ_0 among its quadratic twists. Second, Theorem D fixes ρ_0 against ρ_0'': modulo 2 the system of $\rho_0'' \otimes \chi^3$ has Serre conductor 1951^2, so it cannot occur in $S_2(\Gamma_0(1951), \chi)$ (cf. Table 1).
- Let g generate tame inertia at 1951. Since ρ has conductor 1951, $\rho(g)$ has eigenvalues $\{1, \chi(g)\}$, so $\rho_0(g)$ has eigenvalues $\{\chi(g)^{-3}, \chi(g)^{-2}\}$ and $\rho_0''(g)$ has eigenvalues $\{\chi(g)^{\pm 1}\}$.
  - Hence $\rho_0'' \otimes \chi^3$ has conductor 1951^2, while $\rho_0'' \otimes \chi$ has conductor 1951 and determinant χ^2.
  - Twisting by a non-trivial quadratic character raises the conductor.
  - So ρ is determined up to isomorphism by M, by $\det \rho = \chi$ and by its conductor.
- $\bar\rho = \rho_0 \bmod \lambda \colon G_{\mathbb{Q}} \to \mathrm{SL}_2(\mathbb{F}_5)$, where $\lambda = (1-\zeta_5)$ is the prime of $\mathbb{Z}[\zeta_5]$ above 5, so $\lambda^2 = (\sqrt{5})$. Its traces are $t_p \bmod \sqrt{5}$, with $\varphi \equiv 3$, and we call it the reduction modulo $\sqrt{5}$. Reduction is injective on $\mathrm{SL}(2,5)$, since its only non-trivial proper normal subgroup is $\{\pm 1\}$, so $\bar\rho$ is surjective, with projective kernel field M. Let ω_5 be the mod-5 cyclotomic character.
- 2 is inert in $\mathbb{Z}[\zeta_5]$, with residue field $\mathbb{F}_{16} = \mathbb{F}_2(z)$, where $z = \zeta_5 \bmod 2$ and $z^4+z^3+z^2+z+1 = 0$. We write $\bar\rho_2 = \rho \bmod 2 \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_{16})$. Its projective image is $\mathrm{SL}_2(\mathbb{F}_4) \cong A_5$, and $\det \bar\rho_2 = \bar\chi$.
- $F = \mathbb{Q}(i, \sqrt{5}) = \mathbb{Q}[y]/(y^4 + 3y^2 + 1)$ and $K = \mathbb{Q}(\sqrt{5}, \sqrt{-3}) = \mathbb{Q}[a]/(a^4 - a^3 + 2a^2 + a + 1)$. Both have class number 1, and the discriminant of K is 225.
- $\delta = -(10 + 2\sqrt{5}) = -4\sqrt{5}\,\varphi$, so that $\mathbb{Q}(\zeta_5) = \mathbb{Q}(\sqrt{5})(\sqrt{\delta})$; indeed $\sqrt{\delta} = 2(\zeta_5 - \zeta_5^{-1})$.
- K_6 is the sextic resolvent field of f_5, defined by $x^6 - 75x^4 - 70x^3 + 826x^2 - 210x - 625$. It is the fixed field of a dihedral subgroup of order 10 in $\mathrm{Gal}(M/\mathbb{Q})$.

3. A conditional reduction

For a number field E, a cuspidal automorphic representation Π of $\mathrm{GL}_2(\mathbb{A}_E)$ *corresponds to* $\rho|_{G_E}$ if, for almost all places v, Π_v is unramified with Satake parameter the eigenvalues of $\rho(\mathrm{Frob}_v)$.

Hypothesis H1. For a number field E: there is a cuspidal Π on $\mathrm{GL}_2(\mathbb{A}_E)$ corresponding to $\rho|_{G_E}$.

Hypothesis H2. For $k \in \{\mathbb{Q}, \mathbb{Q}(\sqrt{5})\}$ and a given cuspidal π on $\mathrm{GL}_2(\mathbb{A}_k)$: there is an isobaric automorphic representation Π' of $\mathrm{GL}_2(\mathbb{A}_{L_5 k})$ whose local component at almost every place w' of $L_5 k$ is the local base change of π_w, where $w' \mid w$.

Theorem A. (i) *Let K' be a quadratic field. Suppose H1 holds for $E = K'$, and H2 holds for $k = \mathbb{Q}$ and the cuspidal representation π of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$ constructed from it in Lemma 3.3. Then ρ is modular, and $L(\rho, s)$ is entire.*

(ii) *The same conclusion holds if, for some quadratic extension E of $\mathbb{Q}(\sqrt{5})$ (for instance F, K or $\mathbb{Q}(\zeta_5)$), H1 holds for E, and H2 holds for $k = \mathbb{Q}(\sqrt{5})$ and for $k = \mathbb{Q}$, for the representations arising in the proof.*

The proof occupies the rest of this section. Throughout, E/k is a quadratic extension with $k \in \{\mathbb{Q}, \mathbb{Q}(\sqrt{5})\}$, and we write $\chi_k = \chi \circ N_{k/\mathbb{Q}}$ and $\varepsilon = \varepsilon_{E/k}$. Since A_5 has no subgroup of index 2, 3 or 4, the field M has no subfield of degree 2, 3 or 4. So $E \cap M = \mathbb{Q}$, as $[E : \mathbb{Q}] \le 4$, and $\rho|_{G_E}$ still has projective image A_5.

Lemma 3.1 (cyclic towers)**.** *Let $\mathbb{Q} = k_0 \subset k_1 \subset \cdots \subset k_r$ be a tower in which each k_{i+1}/k_i is cyclic. Then $\rho|_{G_{k_r}}$ has projective image A_5. The subgroups of A_5 of index at most 5 are A_5 and the conjugates of A_4.*

Proof. The projective image of $G_{k_{i+1}}$ is normal in that of G_{k_i}, with cyclic quotient. Since A_5 is simple and non-abelian, an image equal to A_5 stays A_5. The second statement is standard. □

Remark 3.2. Consequently, base change and descent along cyclic extensions never reduce ρ to a solvable case. An argument of this kind needs at least one non-normal step to a field over which the projective image becomes solvable. The smallest such field is L_5, where the image is A_4. In the octahedral case the corresponding field is the non-normal cubic field used by Tunnell [59] via [32]. In the icosahedral case no cubic or quartic analogue exists.

Lemma 3.3 (central character of the descent)**.** *Suppose Π_E is cuspidal on $\mathrm{GL}_2(\mathbb{A}_E)$ and corresponds to $\rho|_{G_E}$. Then:*

(a) *$\Pi_E = \mathrm{BC}_{E/k}(\Pi_k)$ for a cuspidal Π_k on $\mathrm{GL}_2(\mathbb{A}_k)$ with central character χ_k;*
(b) *for almost all places w of k, $\Pi_{k,w} \cong \rho_w \otimes \mathrm{unr}(s_w)$ with $s_w \in \{\pm 1\}$, and $s_w = 1$ if w splits in E.*

Proof. By strong multiplicity one, Π_E is $\mathrm{Gal}(E/k)$-invariant, because $\rho|_{G_E}$ is the restriction of a representation of G_k. By Langlands [44] it is therefore the base change of a cuspidal Π_k, unique up to $\otimes\varepsilon$. Its central character ω satisfies $\omega \circ N_{E/k} = \chi_k \circ N_{E/k}$, so $\omega \in \{\chi_k, \chi_k\varepsilon\}$.

Suppose $\omega = \chi_k\varepsilon$. Let w be a place of k inert in E, let $\{\alpha, \beta\}$ be the Satake parameters of $\Pi_{k,w}$, and let $\{a, b\}$ be the eigenvalues of $\rho(\mathrm{Frob}_w)$. Local base change gives $\{\alpha^2, \beta^2\} = \{a^2, b^2\}$, and the central character gives $\alpha\beta = -ab$. Hence $\mathrm{Ad}(\Pi_{k,w})$ has parameters $\{-u, 1, -u^{-1}\}$, where $u = a/b$. At split w we have $\mathrm{Ad}(\Pi_{k,w}) = \mathrm{Ad}\,\rho_w$. Comparing Euler factors gives, for a finite set S of places,

$$L^S(s, \mathrm{Ad}\,\Pi_k) = \zeta_k^S(s)\,\frac{L^S(s, \mathrm{Ad}\,\rho|_{G_k} \otimes \varepsilon)}{L^S(s, \varepsilon)}.$$

Indeed, with $X = q_w^{-s}$, both sides equal $[(1-X)(1+uX)(1+u^{-1}X)]^{-1}$ at inert w.

- The left side is holomorphic and non-zero at $s = 1$. This is because $L^S(s, \Pi_k \times \tilde{\Pi}_k) = \zeta_k^S(s) L^S(s, \mathrm{Ad}\,\Pi_k)$ has a simple pole there [33]; see also [23].
- On the right, $\mathrm{Ad}\,\rho|_{G_k} \otimes \varepsilon$ is irreducible and non-trivial, since its image contains A_5. By Brauer induction and Hecke's non-vanishing on $\Re s = 1$, its L-function is holomorphic and non-zero at $s = 1$, as is $L(s, \varepsilon)$. So the right side has a simple pole at $s = 1$.

This contradiction shows $\omega = \chi_k$. The same local computation with $\alpha\beta = ab$ gives $\{\alpha, \beta\} = \pm\{a, b\}$ at inert w, which is (b). □

Lemma 3.4 (the sign is a character)**.** *In the situation of Lemma 3.3, assume H2 for (k, Π_k). Then there is a character μ of $\mathrm{Gal}(E/k)$ such that $\Pi_k \otimes \mu$ corresponds to $\rho|_{G_k}$.*

Proof. Let Π' be the representation of $\mathrm{GL}_2(\mathbb{A}_{L_5k})$ given by H2.

The tetrahedral form. The projective image of $\rho|_{G_{L_5k}}$ is A_4, because $L_5k \cap M = L_5$. By Langlands' theorem for tetrahedral representations over an arbitrary number field [44, §3], there is a cuspidal π_5 on $\mathrm{GL}_2(\mathbb{A}_{L_5k})$ corresponding to $\rho|_{G_{L_5k}}$.

Local comparison. By Lemma 3.3(b) and compatibility of local base change with unramified twists, $\Pi'_{w'} \cong \pi_{5,w'} \otimes \mathrm{unr}(s_w^f)$ at almost every place $w' \mid w$, where f is the residue degree of w'/w.

Identifying Π'. Now base change both Π' and π_5 to the quadratic extension EL_5k/L_5k. The two base changes agree at almost all places:

- above places w' inert in EL_5k, the unramified quadratic twist is killed;
- if w' splits in EL_5k, then Frob_w^f fixes E, so either w splits in E (and $s_w = 1$) or f is even.

The representation $\mathrm{BC}(\pi_5)$ is cuspidal, because $\rho|_{G_{EL_5k}}$ still has projective image A_4. By strong multiplicity one [33], $\mathrm{BC}(\Pi') \cong \mathrm{BC}(\pi_5)$; in particular Π' is cuspidal. By Langlands' description of the fibres of quadratic base change, $\Pi' \cong \pi_5 \otimes (\mu \circ N_{L_5k/k})$ for a character μ of $\mathrm{Gal}(E/k)$. Here we use that $E \cap L_5k = k$, as $[L_5k : k] = 5$ is odd.

Reading off the sign. Let w be a place of k with $\mathrm{tr}\,\rho(\mathrm{Frob}_w) \neq 0$.

- Then Frob_w has projective order 1, 3 or 5. So every $w' \mid w$ has odd residue degree f, and $\operatorname{tr}\rho(\mathrm{Frob}_{w'}) \neq 0$.
- Comparing the two descriptions of $\Pi'_{w'}$ gives $\pi_{5,w'} \cong \pi_{5,w'} \otimes \mathrm{unr}(s_w^f \mu(w)^f)$.
- This forces $s_w^f = \mu(w)^f$. Otherwise the eigenvalues of $\rho(\mathrm{Frob}_{w'})$ would be of the form $\{a, -a\}$, with trace zero.
- As f is odd, $s_w = \mu(w)$.

If $\operatorname{tr}\rho(\mathrm{Frob}_w) = 0$, then $\rho_w \cong \rho_w \otimes \mathrm{unr}(-1)$ and the sign is irrelevant. Hence $(\Pi_k \otimes \mu)_w \cong \rho_w$ for almost all w. □

Proof of Theorem A. For (i), apply Lemmas 3.3 and 3.4 with $(E, k) = (K', \mathbb{Q})$. For (ii), apply them first with $(E, k) = (F, \mathbb{Q}(\sqrt{5}))$ and then with $(\mathbb{Q}(\sqrt{5}), \mathbb{Q})$. Here H2 over $\mathbb{Q}(\sqrt{5})$ is needed because otherwise $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$ could interchange the two descents of Π_F.

In both cases we obtain a cuspidal π on $\mathrm{GL}_2(\mathbb{A}_\mathbb{Q})$ corresponding to ρ, so $L(\rho, s)$ and $L(\pi, s)$ differ by finitely many Euler factors. The Euler factors of $L(\rho, s)$ have no poles in $\Re s > 0$, the inverse Euler factors of $L(\pi, s)$ are entire, and $L(\pi, s)$ is entire. So $L(\rho, s)$ is holomorphic for $\Re s > 0$, and likewise $L(\rho^\vee, s)$, using $\tilde{\pi}$.

For $\Re s < 1$, write the functional equation as

$$L(\rho, s) = W\, N^{1/2-s}\, \frac{L_\infty(\rho^\vee, 1-s)}{L_\infty(\rho, s)}\, L(\rho^\vee, 1-s), \qquad L_\infty(\rho, s) = \Gamma_\mathbb{R}(s+\nu)^2,$$

with $N = 1951$ and $\nu \in \{0, 1\}$ according as $\rho(c) = \pm I$. Here $1/L_\infty(\rho, s)$ is entire, $L_\infty(\rho^\vee, 1-s)$ has no poles in $\Re s < 1$, and $L(\rho^\vee, 1-s)$ is holomorphic there by the first step. □

Many quadratic fields instead of quintic base change. Hypothesis H2 is used only to show that the signs s_w of Lemma 3.3 come from a character. Automorphy over infinitely many imaginary quadratic fields does the same.

Lemma 3.5 (compatibility at all places)**.** *Let E be a number field and let Π be a cuspidal representation of $\mathrm{GL}_2(\mathbb{A}_E)$ corresponding to $\rho|_{G_E}$. Then at every place v of E, Π_v corresponds to $\rho|_{W_{E_v}}$ under the local Langlands correspondence. In particular, at every finite place $v \nmid 1951$, Π_v is unramified with Satake parameter the eigenvalues of $\rho(\mathrm{Frob}_v)$.*

Proof. Let ω be a quasi-character of $E^\times \backslash \mathbb{A}_E^\times$. The last step of the proof of Theorem A, applied over E to $\Pi \otimes \omega$ and $\rho|_{G_E} \otimes \omega$, shows that $L(s, \rho|_{G_E} \otimes \omega)$ is entire. It agrees with $L(s, \Pi \otimes \omega)$ up to finitely many Euler factors, so it has finite order, and by the functional equation and the Phragmén–Lindelöf principle it is bounded in vertical strips.

At every place v, $\rho|_{W_{E_v}}$ is a sum of two characters. Away from 1951 this is clear, since ρ is unramified there and $\rho(c)$ is scalar. At 1951, ρ is tamely ramified, inertia acts through a cyclic group of order 5, and Frobenius acts on it by raising to the power $1951 \equiv 1 \pmod 5$; so the decomposition group has abelian image. Let π_v be the corresponding principal series. By the converse theorem [31, Thm. 12.2], $\otimes_v \pi_v$ is a cuspidal automorphic representation. It agrees with Π at almost all places, so it equals Π by strong multiplicity one. □

Theorem A′. *Let $\mathcal{F}$ be a set of imaginary quadratic fields such that all but finitely many primes split in at least one field of $\mathcal{F}$. Examples are the set of all imaginary quadratic fields, or the fields $\mathbb{Q}(\sqrt{-q})$ for primes $q \equiv 3 \pmod 4$.*

(i) *If H1 holds for every $E \in \mathcal{F}$, then ρ is modular and $L(\rho, s)$ is entire. Hypothesis H2 is not used.*

(ii) *Conversely, if ρ is modular, then H1 holds for every imaginary quadratic field.*

Thus ρ is modular if and only if $\rho|_{G_E}$ is automorphic for every imaginary quadratic field E.

Proof. For $E \in \mathcal{F}$ let Π_E be as in H1. By Lemma 3.5, Π_E corresponds to $\rho|_{G_E}$ at every finite place not above 1951. Fix $E_0 \in \mathcal{F}$ and write $\varepsilon_0 = \varepsilon_{E_0/\mathbb{Q}}$. Let Π be a descent of Π_{E_0} as in Lemma 3.3. Thus $\Pi_p \cong \rho_p \otimes \mathrm{unr}(s_p)$ with $s_p \in \{\pm 1\}$ for all p outside a finite set S_0, which we take to contain 1951 and the primes ramified in E_0, and $s_p = 1$ when p splits in E_0.

Step 1: descents from different fields are twists of each other. Let $E \in \mathcal{F}$ with $E \neq E_0$, let Π^E be a descent of Π_E, and put $L = E_0E$.

- The base changes $\mathrm{BC}_{L/\mathbb{Q}}(\Pi) = \mathrm{BC}_{L/E_0}(\Pi_{E_0})$ and $\mathrm{BC}_{L/\mathbb{Q}}(\Pi^E) = \mathrm{BC}_{L/E}(\Pi_E)$ both correspond to $\rho|_{G_L}$.
- They are cuspidal. The Galois group of the field $\tilde{M}$ of ρ_0 is perfect, so $\tilde{M}$ is linearly disjoint from every multiquadratic field. Hence, for each non-trivial quadratic character η, a set of primes of positive density has $\eta(p) = -1$ and $\operatorname{tr}\rho(\mathrm{Frob}_p) \neq 0$, and therefore $\Pi \ncong \Pi \otimes \eta$ and $\Pi^E \ncong \Pi^E \otimes \eta$.
- By strong multiplicity one the two base changes are isomorphic.
- Langlands' description of the fibres of quadratic base change, applied to L/E (where $\varepsilon_{L/E} = \varepsilon_0 \circ N_{E/\mathbb{Q}}$) and then to $E/\mathbb{Q}$, gives $\Pi^E \cong \Pi \otimes \eta_E$ with η_E in the group generated by ε_0 and ε_E.

Step 2: the sign at primes split in E. Let $p \notin S_0$ be a prime that splits in E and has $\operatorname{tr}\rho(\mathrm{Frob}_p) \neq 0$.

- For $v \mid p$ we have $\Pi^E_p \cong \Pi_{E,v}$, by the local description of base change at split places [44], and $\Pi_{E,v} \cong \rho_p$ by Lemma 3.5.
- On the other hand, η_E is unramified at p, and $\Pi^E_p \cong \Pi_p \otimes \eta_{E,p} \cong \rho_p \otimes \mathrm{unr}(s_p\eta_E(p))$.
- Hence $s_p = \eta_E(p)$. Since $\varepsilon_E(p) = 1$, this equals $\varepsilon_0(p)^{a_E}$ for some $a_E \in \{0,1\}$ that does not depend on p.

Step 3: a_E does not depend on E. Let $E \neq E'$ in $\mathcal{F} \smallsetminus \{E_0\}$. The third quadratic subfield of EE' is real, so $E_0 \not\subset EE'$ and $\mathrm{Gal}(E_0EE'/\mathbb{Q}) \cong (\mathbb{Z}/2)^3$. By Chebotarev in $\tilde{M}E_0EE'$ there are primes as in Step 2 that split in E and E' and are inert in E_0. For them $(-1)^{a_E} = s_p = (-1)^{a_{E'}}$. So $a_E = a$ is the same for all E.

Step 4: conclusion. Let $p \notin S_0$ be a prime that splits in some field of $\mathcal{F}$; by assumption this excludes only finitely many primes.

- If $\operatorname{tr}\rho(\mathrm{Frob}_p) = 0$, then $\rho_p \cong \rho_p \otimes \mathrm{unr}(-1)$ and the sign is irrelevant.
- If p splits in E_0, then $s_p = 1 = \varepsilon_0(p)^a$.
- Otherwise p splits in some $E \neq E_0$ in $\mathcal{F}$, and Step 2 gives $s_p = \varepsilon_0(p)^a$.

Hence $(\Pi \otimes \varepsilon_0^a)_p \cong \rho_p$ for almost all p. So $\Pi \otimes \varepsilon_0^a$ corresponds to ρ, and $L(\rho, s)$ is entire by the last step of the proof of Theorem A.

For the converse, let π correspond to ρ, and let E be imaginary quadratic. Then $\mathrm{BC}_{E/\mathbb{Q}}(\pi)$ is cuspidal, as in Step 1, and it corresponds to $\rho|_{G_E}$ because local base change of unramified representations restricts the Satake parameter. □

Remark 3.6. (1) The argument uses infinitely many fields. For a finite family, Steps 1–3 give no information at the primes inert in all of its fields, and these have positive density. For example, with H1 over E_0, over E and also over the real quadratic subfield E'' of E_0E, the argument gives $s_p = 1$ at the primes split in E_0, $s_p = \varepsilon$ at the other primes split in E, and $s_p = \varepsilon''$ at the other primes split in E''. It cannot relate the signs ε and ε'', and s is a character only if they are equal. Theorem A replaces the infinite family by one field and H2, and Theorem A″ below by two fields. Remark 3.8(2) shows that this example becomes conclusive if the three fields are ramified at a single prime. We have not found Theorem A′ in the literature; its ingredients are Langlands' cyclic base change [44] and strong multiplicity one.

(2) The same argument over $k = \mathbb{Q}(\sqrt{5})$ uses the CM quadratic extensions of $\mathbb{Q}(\sqrt{5})$; for two of them the third quadratic subextension of their compositum is totally real. It shows that $\rho|_{G_{\mathbb{Q}(\sqrt{5})}}$ is automorphic if H1 holds for all of them. Over each of these fields $\bar{\rho}$ is, up to twist, the 5-torsion of elliptic curves (Theorem B and [56]), and over F and K Theorem C provides modular ones. These fields all contain $\sqrt{5}$, so they do not give the final descent from $\mathbb{Q}(\sqrt{5})$ to $\mathbb{Q}$.

(3) *Relation to* $\operatorname{Ad}\rho$. Lemma 3.3 shows that H1 for a single quadratic field already makes $\operatorname{Ad}\rho$ automorphic: it corresponds to $\operatorname{Ad}\Pi_k$ [23]. Booker [4, §2] states that a result of Flicker implies that the modularity of ρ is equivalent to that of $\operatorname{Ad}\rho$. We do not see how to obtain the implication from $\operatorname{Ad}\rho$ to ρ without solving the sign problem of Lemma 3.4.
- A self-dual cuspidal representation of $\mathrm{GL}_3(\mathbb{A}_\mathbb{Q})$ with trivial central character is the adjoint of a cuspidal π on $\mathrm{GL}_2(\mathbb{A}_\mathbb{Q})$; see Ramakrishnan [51] for this statement.
- If $\operatorname{Ad}\pi$ corresponds to $\operatorname{Ad}\rho$, then $\pi_p \cong \rho_p \otimes \mu_p$ with μ_p unramified, at almost all p. After normalising the central character, $\mu_p = \mathrm{unr}(\pm 1)$, which is the situation of Lemma 3.3(b).
- The L-packet of $\mathrm{SL}(2)$ determined by π does not see this sign. For odd ρ the sign is fixed by the Galois representation that Deligne and Serre attach to π [16]; for Maass forms no such representation is available.

If the sign problem can be solved in another way, H1 for a single quadratic field suffices.

One quadratic and one cyclic cubic field. The proof of Theorem A′ loses the sign s_p at the primes inert in every field it uses. A field of odd degree does not lose it: at a prime of odd residue degree the sign is visible, as in the last step of the proof of Lemma 3.4.

Lemma 3.7 (cyclic descent of odd prime degree)**.** *Let C be a cyclic extension of $\mathbb{Q}$ of odd prime degree ℓ, and suppose H1 holds for C, with Π_C as there.*

(a) $\Pi_C = \mathrm{BC}_{C/\mathbb{Q}}(\pi)$ *for a unique cuspidal π on $\mathrm{GL}_2(\mathbb{A}_\mathbb{Q})$ with central character χ.*

(b) *Let $p \neq 1951$ be unramified in C, and let $\{a, b\}$ be the eigenvalues of $\rho(\mathrm{Frob}_p)$. If p splits in C, then $\pi_p \cong \rho_p$. Otherwise π_p is unramified, and its Satake parameter $\{\alpha, \beta\}$ satisfies $\alpha\beta = ab$ and $\{\alpha^\ell, \beta^\ell\} = \{a^\ell, b^\ell\}$.*

Proof. (a) As in the proof of Lemma 3.3, Π_C is $\mathrm{Gal}(C/\mathbb{Q})$-invariant. By [44] it is the base change of a cuspidal π, unique up to twisting by the characters ψ^j of $\mathrm{Gal}(C/\mathbb{Q})$. The central character ω of π satisfies $\omega \circ N_{C/\mathbb{Q}} = \chi \circ N_{C/\mathbb{Q}}$, so $\omega = \chi\psi^j$ for some j. Twisting by ψ^i multiplies ω by ψ^{2i}, and ℓ is odd, so exactly one twist has central character χ.

(b) Let $w \mid p$ have residue degree $f \in \{1, \ell\}$. By Lemma 3.5, $\Pi_{C,w}$ is unramified with Satake parameter the eigenvalues of $\rho(\mathrm{Frob}_p)^f$. If $f = 1$, then $\pi_p \cong \Pi_{C,w} \cong \rho_p$. If $f = \ell$, then $C_w/\mathbb{Q}_p$ is unramified of degree ℓ and $\mathrm{BC}(\pi_p) = \Pi_{C,w}$.
- A twist of the Steinberg representation has such a twist as base change.
- A supercuspidal π_p has supercuspidal base change: by Clifford theory, an irreducible two-dimensional representation of $W_{\mathbb{Q}_p}$ stays irreducible on a normal subgroup of odd prime index.
- So $\pi_p = \pi(\nu_1, \nu_2)$ with $\nu_i \circ N$ unramified. The norm maps the units of C_w onto $\mathbb{Z}_p^\times$, so the ν_i are unramified.

The base change of $\pi(\nu_1, \nu_2)$ has Satake parameter $\{\alpha^\ell, \beta^\ell\}$, where $\{\alpha, \beta\} = \{\nu_1(p), \nu_2(p)\}$, and $\alpha\beta = \chi(p) = ab$. □

Theorem A″. *Let E be a quadratic field and C a cyclic extension of $\mathbb{Q}$ of odd prime degree ℓ, for instance a cyclic cubic field. If H1 holds for E and for C, then ρ is modular and $L(\rho, s)$ is entire. Conversely, if ρ is modular, then H1 holds for E and for C.*

Proof. Let π be as in Lemma 3.7, and let Π^E be a descent of Π_E as in Lemma 3.3.

Step 1: π is a descent of Π_E. The field $K = EC$ is cyclic of degree 2ℓ over $\mathbb{Q}$. Since A_5 is perfect, $K \cap M = \mathbb{Q}$, so $\rho|_{G_K}$ is irreducible.

- $\mathrm{BC}_{K/\mathbb{Q}}(\pi) = \mathrm{BC}_{K/C}(\Pi_C)$ and $\mathrm{BC}_{K/\mathbb{Q}}(\Pi^E) = \mathrm{BC}_{K/E}(\Pi_E)$ both correspond to $\rho|_{G_K}$.
- Both are cuspidal. Otherwise one of them would be an isobaric sum of two Hecke characters, and its partial Rankin–Selberg L-function with its contragredient would have a pole of order at least 2 at $s = 1$ [33]. Up to finitely many Euler factors that L-function is $L(s, \rho \otimes \rho^\vee|_{G_K})$, whose pole at $s = 1$ is simple because $\rho|_{G_K}$ is irreducible.
- By strong multiplicity one [33] the two base changes are isomorphic.
- Langlands' description of the fibres of base change [44], applied to K/E and then to $E/\mathbb{Q}$, gives $\pi \cong \Pi^E \otimes \psi^j \varepsilon^a$, where ψ generates the characters of $\mathrm{Gal}(C/\mathbb{Q})$ and $\varepsilon = \varepsilon_{E/\mathbb{Q}}$. Both sides have central character χ, so $\psi^{2j} = 1$, and $j = 0$ as ℓ is odd.

So π is Π^E or $\Pi^E \otimes \varepsilon$, a descent of Π_E. By Lemma 3.3(b), $\pi_p \cong \rho_p \otimes \mathrm{unr}(s_p)$ with $s_p \in \{\pm 1\}$ for all p outside a finite set S, and $s_p = 1$ when p splits in E.

Step 2: the sign is 1. Enlarge S to contain 1951 and the primes ramified in C. Let $p \notin S$, and let $\{a, b\}$ be the eigenvalues of $\rho(\mathrm{Frob}_p)$.

- If $a + b = 0$, then $\rho_p \cong \rho_p \otimes \mathrm{unr}(-1)$ and there is nothing to prove.
- If p splits in C, then $\pi_p \cong \rho_p$ by Lemma 3.7(b).
- Otherwise the Satake parameter of π_p is $\{s_p a, s_p b\}$, and Lemma 3.7(b) gives $\{s_p a^\ell, s_p b^\ell\} = \{a^\ell, b^\ell\}$, because ℓ is odd. If $s_p = -1$, this forces $b^\ell = -a^\ell$. Then b/a is a root of unity whose order divides 2ℓ but not ℓ, and it is not -1 because $a + b \neq 0$; so its order is 2ℓ. But the order of b/a is the order of the image of Frob_p in the projective image A_5, which has no element of order $2\ell \geq 6$. Hence $s_p = 1$.

So $\pi_p \cong \rho_p$ for all $p \notin S$, i.e. π corresponds to ρ, and $L(\rho, s)$ is entire by the last step of the proof of Theorem A.

For the converse, let π correspond to ρ. Its base changes along the cyclic extensions $E/\mathbb{Q}$ and $C/\mathbb{Q}$ exist [44], are cuspidal by the argument of Step 1, and correspond to $\rho|_{G_E}$ and $\rho|_{G_C}$. □

Remark 3.8. (1) The field C is totally real, because complex conjugation has odd order on it, and $\rho|_{G_C}$ is totally even. So Theorem A″ trades the infinitely many imaginary quadratic fields of Theorem A′ for one totally real field, over which the parity problem is the same as over $\mathbb{Q}$. The quadratic field reduces the ambiguity at each prime to a sign, and the odd residue degrees in C remove it. With $E = \mathbb{Q}(\sqrt{-7})$, where H1 holds numerically (Remark 3.10), the remaining hypothesis is H1 for one cyclic cubic field, for instance $\mathbb{Q}(\zeta_7)^+$.

(2) *Normalised descents.* If a quadratic field E is ramified at a single prime $q \neq 1951$, then exactly one descent of Π_E is unramified at q, and it is unramified at every prime other than 1951.

- At q a descent is not a twist of the Steinberg representation and not supercuspidal, since its base change is unramified. So it is $\pi(\nu_1, \nu_2)$ with ν_i trivial on norms from E_q. As χ is unramified at q, ν_1 and ν_2 agree on $\mathbb{Z}_q^\times$: both are unramified, or both become unramified after twisting by ε_E.
- By Step 1 of the proof of Theorem A′, the normalised descents for two such fields E, E' differ by a character in $\langle \varepsilon_E, \varepsilon_{E'} \rangle$. Every non-trivial element of this group is ramified at the prime ramified in E or in E', where both descents are unramified. So the two descents coincide.

For families of fields of prime discriminant, such as $\mathcal{F}_7$, this shortens Steps 2 and 3 of the proof of Theorem A′. It also gives a variant of Theorem A″ with three quadratic fields: if H1 holds for $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-2})$ and $\mathbb{Q}(\sqrt{2})$, which are ramified only at 2, then ρ is modular, since every odd prime splits in one of them. In the example of Remark 3.6(1), the real field $\mathbb{Q}(\sqrt{qq'})$ is ramified at two primes, and each of its descents may be ramified at one of them.

Among finite families of quadratic fields that are each ramified at a single prime, only those containing the three quadratic subfields of $\mathbb{Q}(\zeta_8)$ cover almost all primes: covering requires an odd number of the quadratic characters to have product 1, and at an odd prime q only one quadratic field is ramified at q alone.

Each of the reductions above that uses finitely many fields involves a field with a real place: the totally real quintic L_5 in Theorem A, the field C in Theorem A″, and $\mathbb{Q}(\sqrt{2})$ in Remark 3.8(2). The next proposition shows that, for arguments of this kind, finitely many fields without real places cannot replace it.

Proposition 3.9 (complex conjugation)**.** *Let $\mathcal{E}$ be a finite set of number fields without real places, and let L be the Galois closure of their compositum. Let P be the set of primes p unramified in L whose Frobenius is conjugate to a complex conjugation in* $\mathrm{Gal}(L/\mathbb{Q})$*. Then P has positive density, and for $p \in P$ every prime above p of every $E \in \mathcal{E}$ has residue degree 2. Consequently, for $p \in P$, the local base changes of ρ_p and of $\rho_p \otimes \mathrm{unr}(-1)$ to every completion of every $E \in \mathcal{E}$ above p coincide.*

Proof. The primes of E above p correspond to the orbits of $\langle \mathrm{Frob}_p \rangle$ on the embeddings of E into L. A complex conjugation has order 2 and fixes no such embedding, because E has no real place; so every orbit has size 2. Base change along an unramified quadratic extension sends a Satake parameter $\{\alpha, \beta\}$ to $\{\alpha^2, \beta^2\}$, which does not change when both are multiplied by -1. The density statement is Chebotarev's theorem. □

So an argument that uses H1 over finitely many fields without real places only through the local components $\Pi_{E,w}$, as the proofs of Theorems A′ and A″ do, cannot distinguish ρ_p from $\rho_p \otimes \mathrm{unr}(-1)$ for $p \in P$. Such arguments need either infinitely many fields without real places, as in Theorem A′, or a field with a real place, over which ρ is again even. For imaginary quadratic fields, the primes left undetermined are exactly those whose Frobenius in the compositum is complex conjugation.

Remark 3.10 (numerical). For an imaginary quadratic field E in which 1951 is inert, compute the coefficients of $L(s, \rho|_{G_E}) = \sum c(\mathfrak{n}) N\mathfrak{n}^{-s}$ from the trace table, and put

$$\Pi(z, r) = \sum_{0 \neq \nu \in \mathcal{O}_E} c(\nu \mathcal{O}_E)\, r\, K_0\Big(\frac{4\pi|\nu| r}{\sqrt{|d_E|}}\Big)\, e\Big(\frac{2\,\mathrm{Im}(\nu z)}{\sqrt{|d_E|}}\Big), \qquad z \in \mathbb{C},\ r > 0.$$

Each term is an eigenfunction of the Laplacian of hyperbolic 3-space with eigenvalue 1, the archimedean type of an Artin representation. If H1 holds for E, then Π is a cusp form for $\Gamma_0(1951\mathcal{O}_E)$ with nebentypus $\chi(N(d))$ (for class number $h > 1$, its component at the trivial class). We compared $\Pi(\gamma P)$ with $\chi(N(d))\Pi(P)$ for 8 random γ with lower-left entry 1951, at points P and γP of height about $1/1951$, summing over $N(\nu) \leq X$:

E	h	X	largest defect for ρ	smallest defect of a control
$\mathbb{Q}(\sqrt{-7})$	1	10^8	$2.9 \cdot 10^{-10}$	$3.3 \cdot 10^{-6}$
$\mathbb{Q}(\sqrt{-23})$	3	$3.3 \cdot 10^8$	$9.6 \cdot 10^{-10}$	$5.3 \cdot 10^{-5}$
$\mathbb{Q}(\sqrt{-31})$	3	$4.4 \cdot 10^8$	$2.4 \cdot 10^{-9}$	$1.5 \cdot 10^{-5}$
$\mathbb{Q}(i)$	1	10^8	$6.4 \cdot 10^{-11}$	$5.0 \cdot 10^{-6}$
$\mathbb{Q}(\sqrt{-2})$	1	$2 \cdot 10^8$	$2.1 \cdot 10^{-10}$	$1.2 \cdot 10^{-6}$

The controls are: a_2 changed by 10^{-3}, a_{1951} replaced by 1, the $\sqrt{5}$-conjugate traces, the traces of ρ_0 instead of ρ, and the conjugate nebentypus. Since H1 follows from the modularity of ρ, these are consistency checks. Over every field the series is unchanged if the signs of a_p are changed at primes inert in E, as Proposition 3.9 predicts at the level of local components; over $\mathbb{Q}$ such changes destroy invariance under $\Gamma_0(1951)$.

Over the real quadratic field $F = \mathbb{Q}(\sqrt{2})$ of Remark 3.8(2), 1951 splits, the narrow class number is 1, and $\rho(c) = -I$ at both real places. If H1 holds for F, the expansion at the cusp ∞ of the newform of $\rho|_{G_F}$ is, up to a constant,

$$f(z_1, z_2) = \sum_{0 \neq \alpha \in \mathbb{Z}[\sqrt{2}]} c(\alpha \mathcal{O}_F)\, \mathrm{sgn}(N\alpha) \sqrt{y_1 y_2}\, K_0\Big(\frac{\pi |\alpha_1| y_1}{\sqrt{2}}\Big) K_0\Big(\frac{\pi |\alpha_2| y_2}{\sqrt{2}}\Big) e\Big(\frac{\alpha_1 x_1 - \alpha_2 x_2}{2\sqrt{2}}\Big)$$

on $\mathbb{H} \times \mathbb{H}$, where α_1, α_2 are the real embeddings of α; each term has eigenvalue $1/4$ for both Laplacians. We summed the $4.7 \cdot 10^8$ terms with $|\alpha_j| y_j \leq 6\sqrt{2}$ and compared $f(\gamma P)$ with $\chi(N(d)) f(P)$ for 6 random γ with lower-left entry 1951, at points of height about $1/1951$ in both factors. The largest relative defect is $5.6 \cdot 10^{-9}$. Changing a_2 by 10^{-3} gives defects between $1.9 \cdot 10^{-3}$ and $1.8 \cdot 10^{-2}$. The other signs at the real places, the $\sqrt{5}$-conjugate traces and the conjugate nebentypus give at least 1.6. So all three fields of Remark 3.8(2) pass this test.

Remark 3.11 (identities from solvable subgroups). Let $\mathbf{4}_f = \mathrm{Sym}^3 \rho_0$ and $\mathbf{6}_f = \mathrm{Sym}^5 \rho_0$, the faithful irreducible representations of $\mathrm{SL}(2,5)$ of dimensions 4 and 6. Let α be a character of order 3 of $\mathrm{Gal}(\tilde{M}/L_5) \cong \mathrm{SL}(2,3)$. A computation with the characters of $\mathrm{SL}(2,5)$ gives

$$\mathrm{Ind}_{K_6}^{\mathbb{Q}}\big(\rho|_{G_{K_6}}\big) \cong \rho \oplus \mathbf{4}_f\chi^3 \oplus \mathbf{6}_f\chi^3, \qquad \mathrm{Ind}_{L_5}^{\mathbb{Q}}\big(\rho|_{G_{L_5}} \otimes \alpha\big) \cong \mathbf{4}_f\chi^3 \oplus \mathbf{6}_f\chi^3.$$

Hence, for every Dirichlet character ψ,

$$L(s, \rho \otimes \psi) = \frac{L\big(s, (\rho \otimes \psi)|_{G_{K_6}}\big)}{L\big(s, (\rho \otimes \psi)|_{G_{L_5}} \otimes \alpha\big)}.$$

The numerator and denominator are L-functions of cuspidal representations over K_6 (dihedral) and over L_5 (tetrahedral), so both are entire.

- Such identities give the meromorphic continuation of $L(\rho, s)$, and its holomorphy and non-vanishing on $\Re s \geq 1$. With the functional equation, as at the end of the proof of Theorem A, $L(\rho, s)$ is also holomorphic on $\Re s \leq 0$, so only the strip $0 < \Re s < 1$ is in question.
- They cannot exclude a pole in the critical strip. The virtual characters $-\rho + \mathbf{6}_f\chi^3$ and $-\rho + \rho_0''\chi^3 + \mathbf{4}_f\chi^3$ restrict to genuine characters on every solvable subgroup of $\mathrm{Gal}(N/\mathbb{Q}) \cong \mathrm{SL}(2,5) \times C_5$, where N is the field cut out by ρ.
- So a pole of $L(\rho, s)$ at a zero of $L(s, \mathbf{6}_f\chi^3)$, or at a common zero of $L(s, \rho_0''\chi^3)$ and $L(s, \mathbf{4}_f\chi^3)$, is compatible with every identity obtained by induction from solvable subgroups. For the first configuration see Booker [4].

This is why some non-solvable input, such as H1 or H2, is needed.

Remark 3.12 (status of the hypotheses). (1) *H1.* Over an imaginary quadratic or CM field the parity obstruction disappears, and residual automorphy is available:

- modulo 2 over every imaginary quadratic field, by Theorem D and base change;
- modulo $\sqrt{5}$ over F and K, via the modular elliptic curves of Theorem C. By Theorem B, no elliptic curve over a quadratic field realises it; this does not exclude residual automorphy over a quadratic field by other means.

What is missing is a modularity lifting theorem in the non-cohomological (Artin) weight over a CM field. We know of no such theorem. Theorem A′ asks for H1 over infinitely many imaginary quadratic fields. The residual input modulo 2 is available for all of them, so the missing step is the same. Theorem A″ asks for H1 over one quadratic field and one cyclic cubic field, which is totally real, so that the parity obstruction returns there.

(2) *H2.* Non-normal base change for GL(2) is known for cubic extensions [32]. Wang [61] has recently proved base change for GL(2) and GL(3) along non-normal quartic extensions with Galois closure A_4. He uses the converse theorem, automorphic induction to GL(4) via Kim's exterior square lift, and cyclic base change. He also notes that further cases would need base

change for GL(4) along non-normal quintic extensions, there with solvable Galois closure (dihedral of order 10, or the Frobenius group of order 20). H2 asks for base change for GL(2) along a non-normal quintic extension whose Galois closure is the non-solvable group A_5.

- When π is known to correspond to a Galois representation, H2 often holds for easier reasons. For π of solvable type it follows from Langlands–Tunnell over L_5k, and for many π attached to elliptic curves over $\mathbb{Q}$ it follows from modularity lifting over the totally real field L_5k. This does not help here: the π in Theorem A is not known to be attached to a Galois representation.
- We know of no case of H2 for π not already known to be of Galois type.

Getz [24] proposes a trace formula approach to non-solvable base change, conditional on conjectural trace identities. Kim's non-normal quintic automorphic induction [40] assumes that the icosahedral representation is automorphic, which is known for odd ones over $\mathbb{Q}$. By Theorem A′, H2 can be traded for H1 over infinitely many imaginary quadratic fields.

(3) *Entanglement.* The converse-theorem route to H2 needs the twists $L(s, \pi \times \mathrm{Ind}_{L_5}^{\mathbb{Q}} \lambda)$ for Hecke characters λ of L_5, and so automorphic induction from L_5 to GL(5). For $\lambda = 1$, $\mathrm{Ind}_{L_5}^{\mathbb{Q}} 1 = 1 \oplus \mathbf{4}$, where $\mathbf{4} \cong \rho_0 \otimes \rho_0''$ is the four-dimensional representation of A_5. For odd icosahedral representations the automorphy of $\mathbf{4}$ follows from that of ρ_0 and ρ_0'' [49], which is how [40] proceeds. For our totally real A_5 field it is open. The case $\lambda = 1$ might be avoided with twists highly ramified at a fixed place, but every other λ needs non-normal quintic automorphic induction, which is equally open. So this route to H2 is entangled with icosahedral automorphy.

4. Where the mod $\sqrt{5}$ reduction can live

Theorem B. *Let K_0 be a number field of degree at most 4. Suppose there are an elliptic curve E'/K_0 and a character $\psi\colon G_{K_0} \to \bar{\mathbb{F}}_5^\times$ with $E'[5] \otimes \bar{\mathbb{F}}_5 \cong \bar\rho|_{G_{K_0}} \otimes \psi$. Then K_0 is totally imaginary and contains $\sqrt{5}$, so it is a CM quartic field. Conversely, if K_0 is a CM quartic field containing $\sqrt{5}$, there is a character $\psi\colon G_{K_0} \to \mathbb{F}_5^\times$ with $\psi^2 = \omega_5|_{G_{K_0}}$.*

Proof. Necessity.

- The Weil pairing gives $\det E'[5] = \omega_5$, and $\det \bar\rho = 1$, so $\psi^2 = \omega_5$.
- At a real place with complex conjugation c, $\omega_5(c) = -1$, while $\psi(c)^2 = \psi(c^2) = 1$. So K_0 is totally imaginary.
- Let $\tilde M$ be the field of ρ_0, with $\mathrm{Gal}(\tilde M/\mathbb{Q}) \cong \mathrm{SL}(2,5)$. Since $\mathrm{SL}(2,5)$ is perfect, $\tilde M \cap \mathbb{Q}(\zeta_5) = \mathbb{Q}$, so $(\bar\rho, \omega_5)$ maps $G_{\mathbb{Q}}$ onto $\mathrm{SL}_2(\mathbb{F}_5) \times \mathbb{F}_5^\times$.
- $\mathrm{SL}_2(\mathbb{F}_5)$ has no proper subgroup of index at most 4. Hence every subgroup of $\mathrm{SL}_2(\mathbb{F}_5) \times \mathbb{F}_5^\times$ of index at most 4 contains $\mathrm{SL}_2(\mathbb{F}_5)$, and $(\bar\rho, \omega_5)$ maps G_{K_0} onto $\mathrm{SL}_2(\mathbb{F}_5) \times \omega_5(G_{K_0})$.
- Suppose $\sqrt{5} \notin K_0$. Then $\omega_5(G_{K_0})$ contains a non-square $c \in \mathbb{F}_5^\times$, so there is $g \in G_{K_0}$ with $\bar\rho(g) = I$ and $\omega_5(g) = c$. Then $\psi(g) \notin \mathbb{F}_5$, and $\mathrm{tr}(\bar\rho \otimes \psi)(g) = 2\psi(g) \notin \mathbb{F}_5$. But the traces of $E'[5]$ lie in $\mathbb{F}_5$, a contradiction.

Sufficiency. If $\zeta_5 \in K_0$, take $\psi = 1$. Otherwise $\omega_5|_{G_{K_0}}$ is the quadratic character of $K_0(\zeta_5) = K_0(\sqrt{\delta})$. A character ψ of order 4 with $\psi^2 = \omega_5$ exists if and only if $K_0(\sqrt{\delta})/K_0$ embeds in a cyclic quartic extension, i.e. if and only if the quaternion algebra $(\delta, -1)$ splits over K_0. Over $\mathbb{Q}(\sqrt{5})$ this algebra behaves as follows:

- It is ramified at both real places, since δ is totally negative.
- It is unramified at finite places not dividing $2\delta = -8\sqrt{5}\varphi$.
- It is split at the place $(\sqrt{5})$, since -1 is a unit there and a square in the residue field $\mathbb{F}_5$.
- By the product formula, it is therefore split at the place above 2 as well.

So it splits over every totally imaginary extension of $\mathbb{Q}(\sqrt{5})$, by the Hasse principle. □

Remark 4.1. For each imaginary quadratic field, Chebotarev provides a single prime certifying the contradiction: p split, $p \equiv \pm 2 \pmod 5$ and $t_p \not\equiv 0 \pmod{\sqrt{5}}$. Here $t_p \equiv 0 \pmod{\sqrt{5}}$ exactly when Frobenius has projective order 2.

- For the 31 fields of discriminant $-d$ with $d \le 100$, the least such prime is at most 53. For example, it is 13 for $\mathbb{Q}(i)$ and 2 for $\mathbb{Q}(\sqrt{-7})$.
- For twelve fields we checked that such primes form a proportion 0.375 ± 0.001 of the split primes $7 \le p < 3.2 \cdot 10^6$, the Chebotarev value $3/8$.
- Over F there are none, automatically: degree-one primes of F lie above $p \equiv \pm 1 \pmod 5$.

Once ψ exists, the genus-zero twisting argument of Shepherd-Barron and Taylor [56] produces elliptic curves E' with $E'[5] \cong \bar\rho|_{G_{K_0}} \otimes \psi$, since $\det(\bar\rho \otimes \psi) = \omega_5$. Combined with Theorem B, this shows that among fields of degree at most 4 such curves exist exactly over the CM quartic fields containing $\sqrt{5}$. We do not rely on this: Theorem C gives explicit curves over F and K.

5. Two explicit curves

Klein's resolvent. The trace form on the lattice of trace-zero elements of $\mathcal{O}_{L_5}$ is isometric to the root lattice A_4 with its form multiplied by 1951 (Lemma 8.1). For $d \in \{1, 3\}$ there are $u, v \in \mathcal{O}_{L_5}$ with

$$\operatorname{Tr} u = \operatorname{Tr} v = \operatorname{Tr} uv = 0, \qquad \operatorname{Tr} u^2 = d \operatorname{Tr} v^2.$$

For $d = 1$ both norms are 3902:

$$u = \tfrac{1}{19454}\big(-229x^4 - 27x^3 + 180034x^2 + 602971x - 653282\big),$$
$$v = \tfrac{1}{68089}\big(689x^4 - 386x^3 - 535218x^2 - 1566799x + 622376\big).$$

For $d = 3$ the norms are 11706 and 3902:

$$u = \tfrac{1}{136178}\big(3275x^4 - 1143x^3 - 2565972x^2 - 7306095x + 8710406\big),$$
$$v = \tfrac{1}{19454}\big(-145x^4 - 187x^3 + 112806x^2 + 519501x + 192652\big).$$

Then $y = u + \sqrt{-d}\,v$ satisfies $\operatorname{Tr} y = \operatorname{Tr} y^2 = 0$, so it gives a principal quintic over $\mathbb{Q}(\sqrt{-d})$. The square root $\sqrt{-d}$ is Klein's accessory irrationality: since L_5 is totally real, $\operatorname{Tr} y^2 = 0$ cannot be achieved over a real field.

Let $y_0, \dots, y_4$ be the conjugates of y, labelled so that a 5-cycle in $\mathrm{Gal}(M/\mathbb{Q})$ acts by $\nu \mapsto \nu + 1$. The coordinates $c_k = \sum_\nu \zeta_5^{-k\nu} y_\nu$ then satisfy $c_0 = 0$ and $c_1c_4 + c_2c_3 = 0$. On the ruling $z = c_2/c_4$ of this quadric, A_5 acts by icosahedral substitutions [42]. Let f and H be the forms vanishing on the orbits of size 12 and 20, and put $j = 1728\,J$, where $J = \kappa H^3/f^5$ with the constant κ chosen so that $J = 1$ on the orbit of size 30. Thus $j = 0$, 1728, ∞ on the orbits of size 20, 30, 12, and j is the quotient map $X(5) \cong \mathbb{P}^1 \to X(1)$. The value $j(z)$ is Klein's invariant of y; it lies in $\mathbb{Q}(\sqrt{5}, \sqrt{-d})$.

At 600 digits we obtain the following.

- For $d = 1$: $j_0 = \alpha + \beta\sqrt{5} \in F$ with $\alpha, \beta \in \mathbb{Q}(i)$, with common denominator $29^5 \cdot 41^5 \cdot 1951^2 \cdot 348500149^5 \cdot 696201529^5$.
- For $d = 3$: $j_1 = \alpha' + \beta'\sqrt{5} \in K$ with $\alpha', \beta' \in \mathbb{Q}(\sqrt{-3})$, with common denominator $31^5 \cdot 1951^2 \cdot 22235569^5 \cdot 80579460020178109^5$.

The rational coordinates were recognised with residual below 10^{-590}; the exact values are in the supplement. Theorem C concerns these exact values and does not depend on the recognition step.

Theorem C. *Let E_0/F and E_1/K be elliptic curves with $j(E_0) = j_0$ and $j(E_1) = j_1$, and write (E, K_0) for either pair.*

(i) *The field generated over K_0 by a root of $(X^2+250X+3125)^3 - j(E)X^5$, i.e. the field of definition of a 5-isogeny of E, equals K_0K_6. Consequently the projective representations of $E[5]$ and of $\bar\rho|_{G_{K_0}}$ agree up to conjugation in $\mathrm{PGL}_2(\mathbb{F}_5)$, and $E[5]\otimes\bar{\mathbb{F}}_5 \cong \bar\rho|_{G_{K_0}}\otimes\psi$ for a character ψ.*

(ii) *E has potentially good reduction away from the following places, where $v(j)<0$:*
- *for E_0: one place above each of* 29, 41, 348500149 *and* 696201529 *(with $v=-5$), and the two places above* 1951*;*
- *for E_1: one place above each of* 31, 22235569 *and* 80579460020178109 *(with $v=-5$), and the four places above* 1951*.*

(iii) *E is modular.*

Proof. (i) The fibre polynomial is irreducible over K_0, and its stem field has degree 24 over $\mathbb{Q}$. A computation with `nfisincl` shows that K_0K_6, also of degree 24, embeds in it, so the two fields are equal. Passing to Galois closures over K_0, the projective 5-torsion field of E is K_0M. Since $\sqrt{5}\in K_0$, both projective representations take values in $\mathrm{PSL}_2(\mathbb{F}_5)\cong A_5$. They are surjective maps $G_{K_0}\to A_5$ with the same kernel, so they differ by an automorphism of A_5, and every such automorphism is conjugation by an element of $\mathrm{PGL}_2(\mathbb{F}_5)$. Two lifts of the same projective representation differ by a character.

(ii) The primes dividing the denominators are those listed. Computing $v_P(j)$ at the places P above them gives the list.

(iii) We use the modularity theorem of Caraiani and Newton for elliptic curves over imaginary CM fields [11, §6.1] at $p=5$, the result from which their Corollary 6.1.2 is deduced, in the form stated in version 3 of the preprint. There, E over an imaginary CM field K_0 with $\zeta_5\notin K_0$ is modular if $\bar\rho_{E,5}|_{G_{K_0(\zeta_5)}}$ is absolutely irreducible and $\bar\rho_{E,5}$ is decomposed generic. Decomposed generic means: some prime $\ell\neq 5$ splits completely in K_0, and at every place $v\mid\ell$ the ratio of the eigenvalues of $\bar\rho_{E,5}(\mathrm{Frob}_v)$ is not $\ell^{\pm1}$. These hypotheses hold here:

- K_0 is an imaginary CM field, Galois over $\mathbb{Q}$, and $\zeta_5\notin K_0$.
- The field $K_0(\zeta_5)$ is abelian over $\mathbb{Q}$, so it is linearly disjoint from the field $\tilde M$ of ρ_0. Hence $\bar\rho(G_{K_0(\zeta_5)})=\mathrm{SL}_2(\mathbb{F}_5)$. Twisting by ψ does not change commutators, so the image of $\bar\rho_{E,5}|_{G_{K_0(\zeta_5)}}$ contains $\mathrm{SL}_2(\mathbb{F}_5)$. In particular it is absolutely irreducible.
- Since $\bar\rho_{E,5}\cong\bar\rho|_{G_{K_0}}\otimes\psi$, the eigenvalue ratio at $v\mid\ell$ is that of $\bar\rho(\mathrm{Frob}_\ell)$. For $\ell\equiv 1 \pmod 5$ split in K_0, the condition holds whenever Frob_ℓ has projective order 2 or 3, and such ℓ exist by Chebotarev, as $\tilde M$ and $K_0(\zeta_5)$ are linearly disjoint. Explicit witnesses are also found directly, by computing $a_v(E)\not\equiv\pm(1+\ell) \pmod 5$ at the four places $v\mid\ell$: $\ell=89$ for E_0 and $\ell=19$ for E_1. (Compare Allen–Newton [2, Lemma 2.3].)

□

Remark 5.1 (further checks).
- *Frobenius.* For E_1, the projective order of Frobenius on $E_1[5]$ agrees with that of $\rho(\mathrm{Frob}_p)$ at all 1227 primes of K of good reduction above $p\le 500$, for each of the four conjugates of j_1. Moreover $a_P(E_1)^2\equiv N(P)\,\bar t_p^{\,2} \pmod 5$ at all 383 degree-one primes with $p<3000$, i.e. $\psi(\mathrm{Frob}_P)^2=\omega_5(\mathrm{Frob}_P)$. For E_0 the analogous Frobenius check holds at 970 primes of F.
- *Witnesses.* Decomposed-generic witnesses can also be exhibited directly.
  - For $E_0[5]$ there are 30 among the 41 testable split primes below 1200, the smallest being 89. There are 44 split primes in that range: 29 and 41 have a place with $v(j_0)=-5$, and 1069 has a place where $j_0\equiv 0$.
  - For $E_1[5]$ there are 34 among the 51 split primes below 1500 with good reduction at all four places, the smallest being 19.

- *Genericity rule.* At a split prime ℓ the condition is that the ratio of the Frobenius eigenvalues is not $\ell^{\pm 1}$. For $\ell \equiv 1 \pmod 5$ this excludes Frobenius of projective order 1 and 5; for $\ell \equiv 4 \pmod 5$ it excludes projective order 2.
- *The norm conic.* A point of the conic $x^2 + z^2 = \delta$ over K, such as the one in the supplement, certifies only the existence of ψ. It is not a point of the twisted modular curve.
- *Integral models.* Integral Weierstrass models of E_0 and E_1 are in the supplement.

6. The reduction modulo 2

Let $S = S_2(\Gamma_0(1951), \chi)$, which has dimension 161. Since χ is primitive, every form in S is new. Let $S_{\mathcal{O}}$ be the lattice of forms in S with q-expansion in $\mathcal{O} = W(\mathbb{F}_{16}) = \mathbb{Z}_2[\zeta_5]$, let $\mathbb{T}$ be the $\mathcal{O}$-algebra generated by the Hecke operators acting on it, and let $\bar{S} = S_{\mathcal{O}} \otimes \mathbb{F}_{16}$. The trivial-character space $S_2(\Gamma_0(1951))$, of dimension 162, cannot contain $\bar{\rho}_2$, because $\det \bar{\rho}_2 = \bar{\chi} \neq 1$.

The representation $\bar{\rho}_2$ is unramified at 2, and $\bar{\rho}_2(\mathrm{Frob}_2)$ has the distinct eigenvalues

$$x = z + z^2, \qquad y = 1 + z^3,$$

so that $x + y = z^4 = \bar{a}_2$ and $xy = z^3 = \bar{\chi}(2)$. In weight 2 and level prime to 2, $T_2 \equiv U_2 \pmod 2$ on q-expansions; below, U_2 on $\bar{S}$ means T_2 modulo 2.

Theorem D. (i) *There are normalised forms $G_x, G_y \in \bar{S}$ with $T_\ell G = \bar{a}_\ell G$ for every prime $\ell \nmid 2 \cdot 1951$, and with $U_2 G_x = x G_x$, $U_2 G_y = y G_y$.*

(ii) *The joint eigenspace of T_3, T_5, T_7 in $\bar{S}$ at $(\bar{a}_3, \bar{a}_5, \bar{a}_7)$ is $\mathbb{F}_{16} G_x \oplus \mathbb{F}_{16} G_y$. In particular each of the two $\bar{\rho}_2$-eigenspaces is one-dimensional.*

(iii) *The corresponding generalised eigenspace of T_2, T_3, T_5, T_7 has dimension 9: dimension 3 for T_2-eigenvalue x and 6 for y. Equivalently, the localisations of $\mathbb{T}$ at the two maximal ideals attached to $\bar{\rho}_2$ have ranks 3 and 6 over $\mathcal{O}$. After fixing an embedding $\bar{\mathbb{Q}} \hookrightarrow \bar{\mathbb{Q}}_2$, exactly nine newforms in S are congruent to ρ modulo the maximal ideal of $\bar{\mathbb{Z}}_2$: three with $a_2 \equiv x$ and six with $a_2 \equiv y$.*

(iv) *Let g be a holomorphic Hecke eigenform of any weight and level, and let $\mathcal{O}'$ be the ring of integers of a finite extension of $\mathbb{Q}_2(\zeta_5)$ containing its eigenvalues, with maximal ideal $\mathfrak{m}$. Then $a_p(g) \not\equiv a_p \pmod{2\mathfrak{m}}$ for a set of primes p of positive density.*

(v) *There is a Katz cusp form g_1 of weight 1 on $\Gamma_1(1951)$ over $\mathbb{F}_{16}$, on which the diamond operators act through $\bar{\chi}$, with $a_1(g_1) = 1$ and the following properties.*

- *$T_\ell g_1 = \bar{a}_\ell g_1$ for every prime $\ell \nmid 2 \cdot 1951$, and $a_{2n}(g_1) + \bar{\chi}(2)\, a_{n/2}(g_1) = \bar{a}_2\, a_n(g_1)$ for all $n \geq 1$, where $a_{n/2} = 0$ for odd n.*
- *$G_x = A g_1 + y\, V(g_1)$ and $G_y = A g_1 + x\, V(g_1)$. Here A is the Hasse invariant, and $V(g_1) = (\sigma^{-1} g_1)^{\otimes 2}$ is the form of weight 2 with q-expansion $g_1(q^2)$, where $\sigma^{-1} g_1$ is the base change of g_1 along the inverse of the Frobenius σ of $\mathbb{F}_{16}$.*
- *No eigenform of weight one in characteristic 0, of any level and character, has eigenvalues congruent to $\bar{a}_\ell$ modulo a prime above 2 for almost all primes ℓ. In particular, g_1 is not the reduction of a form of weight one and level $\Gamma_1(1951)$ in characteristic 0.*

Proof. The lattice. The PARI basis of S has $\mathbb{Z}[\zeta_5]$-integral q-expansions, but its reduction modulo 2 has rank only 153. We saturate the lattice by repeatedly dividing by 2 the integral combinations that vanish modulo 2. This is legitimate by Sturm's bound [57], which is $2 \cdot 1952/12 < 326$ in weight 2. Four rounds give rank 161.

(i), first part. The predicted q-expansions are

$$\sum \bar{a}_n q^n, \qquad \bar{a}_{2^j} = x^j \text{ or } y^j,$$

with $\bar{a}_n$ multiplicative and given at odd primes $\ell \neq 1951$ by the weight-2 recursion modulo 2. Both lie in $\bar{S}$ for $n \leq 2400$, $n \neq 1951$, i.e. some $G_x, G_y \in \bar{S}$ have these coefficients. Controls with the $\sqrt{5}$-conjugate traces, or with traces shifted by one prime, do not lie in $\bar{S}$. The forms

$T_\ell G - \bar{a}_\ell G$ for $\ell \in \{3,5,7\}$ and $G \in \{G_x, G_y\}$, and $T_2 G_x - x G_x$, $T_2 G_y - y G_y$, lie in $\bar{S}$ and vanish for $n \le 2400/\ell$. Since $2400/7 > 326$, they vanish identically.

(ii) and (iii). The matrices of T_2, T_3, T_5, T_7 on $\bar{S}$ are determined by the coefficients $n \le 326$. They reproduce all coefficients up to $2400/\ell$, and they commute. Kernel computations give:

- dimensions 4, 3, 2 for the joint eigenspaces of T_3, of $\{T_3, T_5\}$ and of $\{T_3, T_5, T_7\}$;
- dimension 1 after adding $T_2 - x$ or $T_2 - y$;
- generalised eigenspaces of dimensions 3 and 6.

Since these kernel dimensions do not change over $\bar{\mathbb{F}}_2$, no maximal ideal of $\mathbb{T}$ other than those of G_x and G_y has the same eigenvalues for $T_2, \dots, T_7$. So the generalised eigenspaces are the full localisations $\bar{S}_\mathfrak{m}$. By the q-expansion duality $S_\mathcal{O} \cong \mathrm{Hom}_\mathcal{O}(\mathbb{T}, \mathcal{O})$, $\dim \bar{S}_\mathfrak{m} = \mathrm{rank}_\mathcal{O} \mathbb{T}_\mathfrak{m}$. Since S consists of newforms, this rank is the number of eigenforms in $S \otimes \bar{\mathbb{Q}}_2$ whose eigenvalues reduce to $\mathfrak{m}$.

(i), all ℓ.

- Every T_ℓ preserves $E' = \mathbb{F}_{16} G_x \oplus \mathbb{F}_{16} G_y$ and commutes with $T_2|_{E'} = \mathrm{diag}(x, y)$. Since $x \ne y$, both G_x and G_y are T_ℓ-eigenforms.
- Their odd coefficients agree everywhere. To see this, let $W = G_x - G_y$ and lift it to $\tilde{W} \in S_\mathcal{O}$.
  - Let $\mathcal{E} = (E_2(\tau) - 1951 E_2(1951\tau))/(1 - 1951) \in M_2(\Gamma_0(1951))$. Its q-expansion is $1 + \frac{4}{325} \sum_{n \ge 1} \big(\sigma(n) - 1951\, \sigma(n/1951)\big) q^n$, so $\mathcal{E} \equiv 1 \pmod 4$.
  - Let τ_0 be the automorphism $\zeta_5 \mapsto \zeta_5^3$ of $\mathcal{O}$, which lifts the inverse Frobenius of $\mathbb{F}_{16}$. Then $\tau_0(T_2 \tilde{W})$ lies in $S_2(\Gamma_0(1951), \chi^3)$.
  - Hence $\Phi = \mathcal{E}\tilde{W} - \tau_0(T_2 \tilde{W})^2$ lies in $S_4(\Gamma_0(1951), \chi)$ and has $\mathcal{O}$-integral coefficients.
  - Modulo 2, Φ has q-expansion $\sum_n w_n q^n - \sum_n w_{2n} q^{2n}$, the odd part of W. It vanishes for all odd $n \le 2400$ with $n \ne 1951$, in particular for $n \le 651$, beyond the weight-4 Sturm bound $4 \cdot 1952/12 < 651$. So $\Phi \equiv 0$.
- By Khare–Wintenberger and Kisin [39, 41] (Serre's conjecture for odd conductor, including $p = 2$), $\bar{\rho}_2$ is the reduction of the Galois representation of a newform h of weight 2 and level $\Gamma_1(1951)$ whose character reduces to $\bar{\chi}$. That character is χ or $\chi\varepsilon_{1951}$, where ε_{1951} is the quadratic character of conductor 1951. As ε_{1951} is odd, $S_2(\Gamma_0(1951), \chi\varepsilon_{1951}) = 0$, so $h \in S$. The reduction of h is an eigenform for all T_ℓ (including T_2) in $\bar{S} \otimes \bar{\mathbb{F}}_2$ with eigenvalues $\bar{a}_\ell$ for $\ell \nmid 2 \cdot 1951$. So it lies in $E' \otimes \bar{\mathbb{F}}_2$ and is a multiple of G_x or of G_y.
- So one of G_x, G_y has eigenvalues $\bar{a}_\ell$ for all $\ell \nmid 2 \cdot 1951$. As their odd coefficients agree, both do.

(iv). Both ρ_g and ρ modulo $2\mathfrak{m}$ have finite image. So the set of $\sigma \in G_\mathbb{Q}$ where their traces differ is a union of conjugacy classes in a finite quotient. It contains complex conjugation:

- $\mathrm{tr}\, \rho_g(c) = 0$, since ρ_g is odd (also when g is an Eisenstein series);
- $\mathrm{tr}\, \rho(c) = \pm 2$, and $2 \notin 2\mathfrak{m}$.

Apply Chebotarev.

(v), the form. For odd m let $b_m = a_m(G_x) = a_m(G_y)$, which is legitimate by the proof of (i). Since $U_2 G_x = x G_x$ and $U_2 G_y = y G_y$, we have $a_{2^j m}(G_x) = x^j b_m$ and $a_{2^j m}(G_y) = y^j b_m$. Put

$$H = \frac{x G_x + y G_y}{x + y}, \qquad h_j = \frac{x^{j+1} + y^{j+1}}{x + y} \quad (j \ge 0), \qquad h_{-1} = 0.$$

Then $a_{2^j m}(H) = h_j b_m$ and $a_{2^j m}(W) = (x + y) h_{j-1} b_m$, so $W(q) = (x + y) H(q^2)$, where $W = G_x - G_y$ as in the proof of (i).

- Let W' be the base change of $W/(x + y)$ along σ. It is a Katz form of weight 2 on which the diamond operators act through $\sigma \circ \bar{\chi} = \bar{\chi}^2$, and its q-expansion is $\sum_k \sigma(a_k(H)) q^{2k} = H(q)^2$.

- The Hasse invariant A has weight 1 and q-expansion 1 [38, §2]. So $H^{\otimes 2}$ and A^2W' are forms of weight 4 with the same q-expansion. By the q-expansion principle [38, §1.6] on the geometrically irreducible curve $X_1(1951)_{\mathbb{F}_{16}}$, they are equal.
- Comparing divisors, $2\operatorname{div} H = 2\operatorname{div} A + \operatorname{div} W' \geq 2\operatorname{div} A$. So $g_1 = H/A$ is a section of ω, i.e. a Katz form of weight 1. It vanishes at the cusps, where A is invertible, and its q-expansion is $H(q)$.

Since $h_j + y\,h_{j-1} = x^j$ and $h_j + x\,h_{j-1} = y^j$, the form $Ag_1 + yV(g_1)$ has the same q-expansion as G_x, and $Ag_1 + xV(g_1)$ the same as G_y. For $\ell \nmid 2 \cdot 1951$, the q-expansion formulas for T_ℓ in weights 1 and 2 agree modulo 2, because ℓ is odd. So $T_\ell g_1$ and $\bar{a}_\ell g_1$ have the same q-expansion, as $T_\ell H = \bar{a}_\ell H$. The identity at 2 is $h_{j+1} + xy\,h_{j-1} = (x+y)h_j$, with $xy = \bar{\chi}(2)$ and $x + y = \bar{a}_2$.

(v), non-liftability. Let f be an eigenform of weight one in characteristic 0 whose eigenvalues are congruent to $\bar{a}_\ell$ for almost all ℓ.

- By Deligne and Serre [16], ρ_f is an odd Artin representation. Its reduction is isomorphic to $\bar{\rho}_2$ by Chebotarev and Brauer–Nesbitt, because $\bar{\rho}_2$ is irreducible.
- So the projective image of ρ_f, a finite subgroup of $\mathrm{PGL}_2(\mathbb{C})$, has the projective image A_5 of $\bar{\rho}_2$ as a quotient. Cyclic, dihedral, tetrahedral and octahedral groups have no such quotient, so the projective image of ρ_f is A_5, and ρ_f has the same projective kernel field M as $\bar{\rho}_2$.
- As M is totally real, $\rho_f(c)$ is scalar. This contradicts $\det \rho_f(c) = -1$.

If g_1 were the reduction of a form of weight one and level $\Gamma_1(1951)$ in characteristic 0, the lemma of Deligne and Serre [16, Lemme 6.11], applied to the Hecke and diamond operators, would produce such an f. □

Remark 6.1 (an encoding-free check). This check works in characteristic 0. Let $D_\ell = \det(a_\ell - T_\ell \mid S)$ for odd $\ell \neq 1951$, and $D_2 = \det(T_2^2 - a_2 T_2 + \chi(2) \mid S)$. These are elements of $\mathbb{Z}[\zeta_5]$. We computed them from PARI's Hecke matrices via complex embeddings at 600 digits, and checked them exactly modulo two primes $\equiv 1 \pmod 5$. If $\bar{\rho}_2$ occurs in S, then $2 \mid D_\ell$ for all ℓ. Table 1 records $v_2(D_\ell)$, together with three controls in which $\bar{\rho}_2$ should not occur.

TABLE 1. $v_2(D_\ell)$ for $S = S_2(\Gamma_0(1951), \chi)$, and controls: the trivial-character space; the $\sqrt{5}$-conjugate traces $\sigma(t_\ell)\chi(\ell)^3$ (the traces of $\rho_0'' \otimes \chi^3$); and traces shifted to the next prime.

ℓ	2	3	5	7	11	13	17	19	23	29	31	37
S	6	7	12	6	6	6	8	6	6	7	7	6
trivial character	0	0	0	10	0	0	0	0				
$\sqrt{5}$-conjugate	6	0	12	6	0	6	0	0				
shifted	0	0	12	0	0	0	0	0				

The $\sqrt{5}$-conjugate control differs from the true system exactly at the primes whose Frobenius has order 5 ($\ell = 3, 11, 17, 19$). The shifted control agrees with it only at $\ell = 5$, where the shifted trace coincides with the true one.

Remark 6.2 (on Theorem D(v)). (1) Theorem D(v) is an instance at $p = 2$ of Gross's picture [28], in which a weight-one form modulo p for a representation unramified at p corresponds to two p-stabilisations in weight p; here it is proved directly. Edixhoven [18, Prop. 4.2] shows that a Katz form of weight 2 modulo 2 whose odd coefficients vanish comes from weight one by Frobenius. Applied to W, this gives a second proof that g_1 exists. As a check on the bookkeeping, $H(q)^2$ lies in the reduction of $S_2(\Gamma_0(1951), \chi^2)$ to $n \leq 2400$, while $H(q)$, and the same construction with $\sqrt{5}$-conjugate traces, do not.

(2) Weight-one forms modulo 2 that do not lift to characteristic 0 are known.
- Mestre's examples of prime level [18, App. A] have image $\mathrm{SL}_2(\mathbb{F}_8)$, which is not a quotient of any finite subgroup of $\mathrm{GL}_2(\mathbb{C})$.
- Wiese [64] obtains non-liftable forms from dihedral representations induced from real quadratic fields, for which no odd lift to characteristic 0 has conductor dividing the level.
- In Theorem D(v) the image is A_5 and the obstruction is parity: $\bar\rho_2$ has the even Artin lift ρ, and no odd Artin lift of any conductor.

(3) The representation ρ lifts $\bar\rho_2$, is unramified at 2 and is minimally ramified at 1951, but by (v) no weight-one form realises ρ or any other Artin lift of $\bar\rho_2$. So a modularity lifting theorem in weight one at $p = 2$, of the kind proved by Calegari and Geraghty for odd p [10], cannot apply to $\bar\rho_2$ without a parity condition, and such a condition excludes ρ.

Remark 6.3 (the $\sqrt{5}$-conjugate). Let σ be the automorphism $\zeta_5 \mapsto \zeta_5^2$ of $\mathbb{Q}(\zeta_5)$, so that $\sigma(\sqrt{5}) = -\sqrt{5}$, and let $\rho'' = \sigma\circ\rho = \rho_0''\otimes\chi$. It has conductor 1951, determinant χ^2 and the same projective kernel field M (§2).

- Theorems A and A′ hold for ρ'' with the same proofs. They use only the projective image, the determinant, the quintic field L_5 and the tameness at 1951, and these are of the same kind for ρ and ρ''.
- The automorphism σ induces the identity on $\mathbb{Z}[\zeta_5]/\lambda = \mathbb{F}_5$, so $\rho_0'' \bmod \lambda \cong \bar\rho$ by Brauer–Nesbitt. Theorems B and C therefore hold verbatim for ρ_0'', with the same curves. A lifting argument starting from E_0 or E_1 would have to distinguish the two lifts ρ_0 and ρ_0'' of $\bar\rho$, whose inertial types at 1951 differ.
- Modulo 2, σ induces the Frobenius of $\mathbb{F}_{16}$. The analogue of Theorem D for ρ'' holds in $S_2(\Gamma_0(1951), \chi^2)$, with U_2-eigenvalues $x^2 = 1 + z + z^3$ and $y^2 = 1 + z$ and generalised eigenspaces of dimensions 3 and 6. We also checked this directly in that space. The base change σg_1 is the corresponding weight-one form, and it does not lift for the same reason.
- We do not know how to deduce the modularity of ρ'' from that of ρ, or conversely.
  - Applying σ to Dirichlet coefficients does not preserve analytic properties. For instance, let ψ be the character modulo 11 with $\psi(2) = \zeta_5$, so that $\sigma\circ\psi = \psi^2$. The first zeros of $L(s,\psi)$ and $L(s,\psi^2)$ on the critical line in the upper half-plane are at heights 5.134 and 4.629.
  - Galois conjugates of Maass forms of eigenvalue 1/4 are not known to be automorphic [50].
  - For odd icosahedral representations over $\mathbb{Q}$ the implication holds, by the rationality of weight-one forms [62]. In the even case Booker [5] and Ramakrishnan [50] take the existence of the conjugate form as a hypothesis.
  - Since $\rho\otimes\rho'' \cong \mathbf{4}\otimes\chi^{-1}$, the route to $\mathbf{4}$ through [49] in Remark 3.12(3) needs the modularity of both ρ and ρ''.

7. Two further computations

This section records two computations made after the results above. They do not bear on the hypotheses of Theorems A and A′, and nothing in them proves that $L(\rho, s)$ is entire.

Weil's quadratic form from the primes. Write $\Lambda(\rho,s) = 1951^{s/2}\Gamma_{\mathbb{R}}(s+1)^2 L(\rho,s)$, and let $\Lambda_\rho(p^k) = \operatorname{tr}\rho(\mathrm{Frob}_p^k)\log p$ for $p \neq 1951$ and $\Lambda_\rho(1951^k) = a_{1951}^k \log 1951$, where a_{1951} is the

eigenvalue of Frobenius on the inertia invariants. For a test function g put

$$W(g) = (\log 1951 - 2\log\pi - 2\gamma_{\mathrm{E}})\, g(0) + 2\int_0^\infty \frac{2e^{-2x}g(0) - e^{-3x/2}\big(g(x)+g(-x)\big)}{1-e^{-2x}}\,dx \\ - \sum_{n\geq 2} \frac{\Lambda_\rho(n) g(\log n) + \overline{\Lambda_\rho(n)}\, g(-\log n)}{\sqrt{n}},$$

the distribution of the explicit formula of $L(\rho, s)$ (γ_{E} is Euler's constant). Let $L = \log X$, let V_m be the span of $e_j(x) = e^{2\pi i j x/L}$ ($|j| \leq m$) on $I = [-L/2, L/2]$, and for $\varphi \in V_m$ put $Q_X(\varphi) = W(\varphi * \tilde{\varphi})$, where $\tilde{\varphi}(x) = \overline{\varphi(-x)}$. Since $\varphi * \tilde{\varphi}$ is supported in $[-L, L]$, only $n \leq X$ enter, and Q_X is defined without any assumption on the analytic behaviour of $L(\rho, s)$. If $L(\rho, s)$ is entire, then $Q_X(\varphi) = \sum_\gamma |\hat{\varphi}(\gamma)|^2$ over the zeros $\frac{1}{2} + i\gamma$; a pole would enter with the opposite sign. Weil's criterion [63] relates positivity for all supports to the Riemann hypothesis for $L(\rho, s)$ together with its holomorphy.

Proposition 7.1 (computer-assisted)**.** *For $(X, m) = (10^4, 30)$, $(10^5, 45)$ and $(10^6, 60)$, the form Q_X is positive definite on V_m: $Q_X(\varphi) \geq \mu_X \|\varphi\|^2_{L^2(I)}$ with $\mu_X = 3.0 \cdot 10^{-19}$, $5.1 \cdot 10^{-45}$ and $1.0 \cdot 10^{-77}$ respectively. For $X = 10^5$ the minimum is at most $5.19 \cdot 10^{-45}$.*

Proof. In the basis (e_j) the archimedean entries have closed forms in terms of ψ and ψ' at $\frac{3}{4} + \frac{\pi i j}{L}$ plus exponentially small sums, and the prime entries are finite sums with $\Lambda_\rho(n)/\log p \in \mathbb{Z}[\zeta_5]$ taken from the trace table of §2. An LDL^{T} factorisation of $Q_X - \mu_X$ Gram in ball arithmetic (Arb [35], at 300, 1600 and 3000 bits) has all pivots certified positive. The upper bound is the Rayleigh quotient of a rounded eigenvector, also in ball arithmetic. □

Remark 7.2 (numerical). (1) The real zeros of the Fourier transform of the eigenvector of the smallest eigenvalue coincide with the zeros of $L(\rho, s)$ computed by PARI [47], which assumes holomorphy: for $X = 10^6$, all 57 zeros with $|\gamma| \leq 21.8$, the first ($\gamma = -1.0364181\ldots$) to within $5 \cdot 10^{-79}$ and the others to within $2 \cdot 10^{-41}$; for $X = 10^5$ and 10^4, the 49 and 40 zeros with $|\gamma| \leq 19.6$ and 16.4. This is the phenomenon Connes [13] observed for $\zeta(s)$ with the primes below 13.

(2) With the traces t_p permuted at random among the primes, Q_{10^4} has four negative eigenvalues, the smallest -3.86, and no zeros are matched.

(3) Adding by hand the term of a simple pole at $\frac{1}{2} + 7.6i$ to Q_{10^5} makes the smallest eigenvalue -7.9. A pole together with an extra zero at distance δ leaves the form indefinite for $\delta \geq 10^{-30}$ and positive for $\delta \leq 10^{-40}$.

Positivity on a finite-dimensional space at one scale follows from holomorphy and the Riemann hypothesis, but implies neither: a pole close enough to a zero is invisible at every finite scale.

p-adic L-functions. Since $\rho(c) = -I$, for an odd prime p the twist $\rho\omega_p$ by the Teichmüller character is totally even, so the p-adic Artin L-function $L_p(s, \rho\omega_p)$ is defined by Brauer induction and Deligne–Ribet [15], with $L_p(1-n, \rho\omega_p) = \det\big(1 - \rho(\mathrm{Frob}_p)p^{n-1}\big) L(\rho, 1-n)$ for $n \equiv 1 \bmod p-1$. By Greenberg [25] and Wiles' proof of the main conjecture [65] (and independently by Ellerbrock and Nickel [19]), $L_p(s, \rho\omega_p)$ has no poles: the p-adic analogue of Artin's conjecture holds for ρ. Moreover its Iwasawa power series in $T = (1+p)^s - 1$ has coefficients in $\mathbb{Z}_p[\zeta_5]$. This is Greenberg's refined conjecture [25, p. 87], which holds for every totally even Artin character when p is odd: it was proved by Ritter and Weiss [52, Remark (G)], see [21, Theorem 3.3], and Greenberg derives it from Wiles' results in [26, §§3–4]. (As a power series, the Euler factor at 1951 has μ-invariant 0, so it does not matter whether it is removed.) Applied to $\rho\omega_p^j$ with $j \equiv n \bmod p-1$, whose value at $1-n$ is $\det(1 - \rho(\mathrm{Frob}_p)p^{n-1})L(\rho, 1-n)$ with an Euler factor $\equiv 1 \bmod p$ when $n \geq 2$, it

shows that $L(\rho, 1-n)$ is integral at every odd prime for odd $n \geq 3$. Values at negative integers carry no information about the critical strip, so none of this bears on $L(\rho, s)$ there.

Lemma 7.3. *Let $G = \mathrm{Gal}(N/\mathbb{Q}) \cong \mathrm{SL}(2,5) \times C_5$, where N is the field cut out by ρ. If $H \leq G$ and $\rho^H \neq 0$, then H is trivial or cyclic of order 5. Consequently, a Hecke L-function over a subfield of N that contains $L(\rho, s)$ as a factor comes from a character whose kernel field has degree at least 120, and the ρ-isotypic part of the S-units of a subfield $K \subset N$ vanishes unless $[K : \mathbb{Q}] \geq 120$. On such an H of order 5 the fixed spaces of ρ, $\rho_0'' \otimes \chi^3$, $\mathbf{4}_f \otimes \chi^3$ and $\mathbf{6}_f \otimes \chi^3$ have dimensions $1, 0, 1, 1$.*

Proof. If ζA, with $\zeta \in \mu_5$ and $A \in \rho_0(G) \cong \mathrm{SL}(2,5)$, fixes $v \neq 0$, then v is an eigenvector of A with eigenvalue ζ^{-1}. The elements of $\mathrm{SL}(2,5)$ with eigenvector v form a finite subgroup of a Borel subgroup of $\mathrm{SL}_2(\mathbb{C})$, hence a cyclic group, and the eigenvalue on v is a faithful character of it, since an element of finite order of $\mathrm{SL}_2(\mathbb{C})$ with both eigenvalues 1 is trivial. So those with a fifth root of unity as eigenvalue on v form a subgroup of order 1 or 5, and ζ is determined by A. If $\rho \subset \mathrm{Ind}_H \psi$ with ψ linear, then $\rho \subset \mathrm{Ind}_{\ker\psi} 1$, so $\rho^{\ker\psi} \neq 0$ by Frobenius reciprocity. The dimensions, and the uniqueness of the class of such subgroups of order 5, were checked in GAP, as was the list of stabilisers of eigenlines of non-central elements (cyclic of orders 4, 6 and 10). □

At a prime p with $\rho(\mathrm{Frob}_p)$ of order 5 and eigenvalue 1, D_p is such a subgroup, $L_p(s, \rho\omega_p)$ vanishes at $s = 0$, and Gross's conjecture [27] predicts a simple zero. By the lemma, every abelian L-function in a Brauer expression of $L(\rho, s)$ that contains $L(\rho, s)$ as a factor comes from an abelian field of degree at least 120; for instance the identity $L(\rho, s) = L(\mathrm{Ind}\,\psi, s)/L(\mathrm{Ind}\,\phi, s)$ with $\mathrm{Ind}\,\psi = \rho + \mathbf{4}_f\chi^3 + \mathbf{6}_f\chi^3$ and $\mathrm{Ind}\,\phi = \mathbf{4}_f\chi^3 + \mathbf{6}_f\chi^3$ uses abelian fields of degrees 120 and 200, with predicted orders 3 and 2.

Remark 7.4 (numerical). From the functional equation, the absolutely convergent Euler product at $s = n$, and the root number $W(\rho) = -\zeta_5^3\tau(\chi)/\sqrt{1951}$ ($\tau(\chi)$ the Gauss sum; selected among Gauss-sum candidates by PARI's numerical root number), we computed $L(\rho, 1-n)$ for 67 odd n with $3 \leq n \leq 1459$, without assuming holomorphy, to absolute accuracy at least 10^{-12} (at least 10^{-46} for $n \geq 11$). By [15] these values lie in $\mathbb{Q}(\zeta_5)$, and by the integrality above they are integral away from 2; numerically each lies in $\mathbb{Z}[\zeta_5]$, for instance $L(\rho, -2) = 8612 - 23244\zeta_5 - 8800\zeta_5^2 + 26804\zeta_5^3$.

(1) For $p = 3, 5, 7, 13$ the values $L_p(1-n, \rho\omega_p)$ are p-adic units, and all 4034 pairwise differences have exactly the valuation predicted for an integral power series in T with unit constant and linear coefficients. By the integrality above, one unit value already makes the power series a unit, so $\lambda_p = \mu_p = 0$ for these p once a single value is known exactly. The values with $n \equiv 1 \bmod (p-1)p^k$ are congruent to $\det(1 - \rho(\mathrm{Frob}_p)) \cdot (-4)(\zeta_5 + \zeta_5^2)$ modulo p^{k+1}, which we checked up to 3^7, 5^4, 7^3 and 13^2; here $-4(\zeta_5 + \zeta_5^2)$ is PARI's value of $L(\rho, 0)$, whose computation assumes that $L(\rho, s)$ is entire, and agrees with it to $7 \cdot 10^{-129}$. For these p the factor $\det(1 - \rho(\mathrm{Frob}_p))$ is a unit, so the integrality above and the interpolation at $s = 0$ [19, Theorem 3] turn the congruences into $L(\rho, 0) \equiv -4(\zeta_5 + \zeta_5^2)$ modulo $3^7 \cdot 5^4 \cdot 7^3 \cdot 13^2$. This uses only the values at $s \leq -2$, as identified above, and so no assumption about the critical strip.

(2) For $p = 11$ and $p = 37$, $\rho(\mathrm{Frob}_p)$ has the eigenvalue 1. At every prime above p the valuations of $L_p(1-n, \rho\omega_p)$, for 22 and 14 values of n, are those of a simple zero at $s = 0$ with unit derivative, the order predicted by Gross.

(3) The cited theorems exclude $p = 2$. With $L^*(n) = \det(1 - \rho(\mathrm{Frob}_2)2^{n-1})\, L(\rho, 1-n)$, every computed $L^*(n)$ is exactly 4 times a 2-adic unit, all of them are $\equiv 4(1 + \zeta_5 + \zeta_5^3) \bmod 8$, and each of the 2211 differences $L^*(n) - L^*(n')$ has valuation exactly $v_2(n - n') + 4$. This is the pattern of $4u(T)$, with $T = 5^s - 1$ and u a unit power series whose linear coefficient is a unit. The factor $4 = 2^{\dim\rho}$ is expected: the 2-adic L-functions of Deligne and Ribet carry a factor

$2^{[F:\mathbb{Q}]}$ [26, (9)], so by Brauer induction the 2-adic L-function of $\rho\omega_4$ (ω_4 the character of $\mathbb{Q}(i)$) is 4 times a quotient of power series, and Greenberg's form of the main conjecture at $p = 2$ [26, §4] predicts that this quotient is integral. That it is a unit is not explained. PARI's value at $s = 0$ fits: with $L(\rho, 0) = -4(\zeta_5 + \zeta_5^2)$ and $\det(1 - \bar{\rho}_2(\mathrm{Frob}_2)) = z + z^2$ in $\mathbb{F}_{16}$, the residue is $(z + z^2)^2 = 1 + z + z^3$ again.

8. The six representations of prime conductor below 10^4

Doud and Moore [17] list all even icosahedral representations of prime conductor below 10^4. Their quintic fields $L_5^{(p)}$ are totally real with Galois closure A_5 and discriminant p^4, for

$$p \in \mathcal{P} = \{1951,\ 2141,\ 3701,\ 3821,\ 8501,\ 9461\}.$$

All these p are $\equiv 1 \pmod 5$, and all but 1951 are $\equiv 5 \pmod 8$. For $p \in \mathcal{P}$ we use the notation of §2 with 1951 replaced by p: χ has order 5 and conductor p, with $\chi(g) = \zeta_5^2$ for the generator g of $(\mathbb{Z}/p)^\times$ chosen by PARI; $\rho = \rho_0 \otimes \chi^3$ has conductor p and determinant χ, where ρ_0 is the one of the two conjugate twist-minimal representations for which this conductor is p; and M, K_6, $\bar{\rho}$, $\bar{\rho}_2$ are as before. Theorems A, A′, A″ and B and their proofs apply to each of them without change. They use only the projective image, the determinant, the field L_5, and tame ramification at $p \equiv 1 \pmod 5$.

Lemma 8.1 (the trace-zero lattice)**.** *Let L be a totally real quintic field of discriminant p^4, where $p > 5$ is prime, and let $\Lambda_0 = \{x \in \mathcal{O}_L : \mathrm{Tr}\, x = 0\}$ with the form $\mathrm{Tr}(xy)$. Then Λ_0 is isometric to the root lattice A_4 with its form multiplied by p.*

Proof.
- *Ramification.* As $p > 5$, the ramification at p is tame, so the exponent of p in the discriminant is $\sum(e_i - 1)f_i$ over the primes above p, where $\sum e_i f_i = 5$. This sum is 4 only for a single prime with $e = 5$.
- *Determinant.* The trace map $\mathcal{O}_L \to \mathbb{Z}$ is onto. Otherwise its image is $5\mathbb{Z}$, then $\mathcal{O}_L = \mathbb{Z} \oplus \Lambda_0$ as an orthogonal sum, and 5 divides the discriminant. So $[\mathcal{O}_L : \mathbb{Z} \oplus \Lambda_0] = 5$, and $5 \det \Lambda_0 = 25p^4$.
- *Divisibility by p.* A totally and tamely ramified extension of $\mathbb{Q}_p$ of degree 5 has a uniformiser π with $\pi^5 = pu$ for a unit $u \in \mathbb{Z}_p^\times$ [43, II, §5, Prop. 12]. So $\mathcal{O}_L \otimes \mathbb{Z}_p = \mathbb{Z}_p[\pi]$, $\mathrm{Tr}\, \pi^k = 0$ for $1 \le k \le 4$, and $\mathrm{Tr}\, \pi^5 = 5pu$. Hence $\Lambda_0 \otimes \mathbb{Z}_p$ has basis $\pi, \dots, \pi^4$, and its Gram matrix is $5pu$ times the anti-diagonal matrix of ones. So $\mathrm{Tr}(xy) \in p\mathbb{Z}$ on Λ_0, and $\Lambda_1 = (\Lambda_0, \mathrm{Tr}(xy)/p)$ is an integral lattice of determinant 5.
- *Evenness.* For $x \in \Lambda_0$, $\mathrm{Tr}(x^2) = \mathrm{Tr}(x)^2 - 2e_2(x)$ is even, where $e_2(x)$ is the second elementary symmetric function of the conjugates. As p is odd, Λ_1 is even.
- *Identification.* Λ_1 is positive definite, since L is totally real. Let $\lambda_1 \le \cdots \le \lambda_4$ be its successive minima (squared lengths). Minkowski's second theorem [12, VIII.2], with Hermite's constant $\gamma_4 = \sqrt{2}$ [14, Ch. 1, Table 1.2], gives $\lambda_1 \cdots \lambda_4 \le \gamma_4^4 \det \Lambda_1 = 20$. The λ_i are even, so they are all 2. Hence the roots of Λ_1 span a root lattice R of rank 4, with $\det R = 5\,[\Lambda_1 : R]^2$. The root lattices of rank 4 have determinants 5 (A_4), 4 (D_4), 8, 9, 12 and 16. So $R = A_4$ and $R = \Lambda_1$.

□

For $p = 1951$ this explains the isometry found by computation in §5; for all $p \in \mathcal{P}$ we also checked it with `qfisom`. It is the analogue, for these non-Galois fields, of the theorem of Mantilla-Soler and Monsurrò that the trace-zero form of a tamely ramified cyclic field of prime degree ℓ is $A_{\ell-1}$ scaled by the radical of the discriminant [45, Thm. 2.9].

Corollary 8.2 (accessory irrationalities)**.** *Let L be as in Lemma 8.1, and let k be a number field without real places. Then $\mathrm{Tr}\, y^2 = 0$ for some $0 \ne y \in \Lambda_0 \otimes k$. For $k = \mathbb{Q}(\sqrt{-d})$ this means*

that $y = u + \sqrt{-d}\,v$ *with* $u, v \in \Lambda_0$, $\operatorname{Tr} uv = 0$ *and* $\operatorname{Tr} u^2 = d \operatorname{Tr} v^2$. *For* $d = 1, 2, 3$ *one can take* $\operatorname{Tr} v^2 = 2p$ *and* $\operatorname{Tr} u^2 = 2dp$.

Proof. By the lemma, $\Lambda_0 \otimes \mathbb{Q}$ is similar to $A_4 \otimes \mathbb{Q} \cong \langle 2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4} \rangle \cong \langle 2, 6, 3, 5 \rangle$.

- A quaternary form over $\mathbb{Q}_\ell$ whose discriminant is not a square is isotropic [55, IV.2.2, Thm. 6]. The discriminant 5 is not a square in $\mathbb{Q}_2$, $\mathbb{Q}_3$, $\mathbb{Q}_5$, nor in $\mathbb{Q}_\ell$ for $\ell \equiv \pm 2 \pmod 5$.
- For $\ell \equiv \pm 1 \pmod 5$ the coefficients are ℓ-adic units, so the Hasse invariant is $1 = (-1, -1)_\ell$, and the form is again isotropic [55, loc. cit.].

So the form is isotropic at every place of k, hence over k by the Hasse–Minkowski theorem. In $A_4 = \{x \in \mathbb{Z}^5 : \sum x_i = 0\}$, the vector $v = e_5 - e_1$ is orthogonal to $e_4 - e_2$, $e_3 + e_4 - e_1 - e_5$ and $2e_4 - e_1 - e_5$, of squared lengths $2d$ for $d = 1, 2, 3$. □

So the first step of Klein's construction in §5 is available over every imaginary quadratic field: any imaginary quadratic irrationality can serve as the accessory irrationality for these quintics. We carried out the remaining steps, at 600 or 700 digits, for $d = 1, 3$ and every $p \in \mathcal{P}$, and for $d = 2$ and $p = 1951$. Let $K_d = \mathbb{Q}(\sqrt{5}, \sqrt{-d})$; then $K_1 = F$, $K_3 = K$, and $K_2 = \mathbb{Q}[a]/(a^4+6a^2+4)$.

Theorem E. *Let* $p \in \mathcal{P}$ *and* $d \in \{1, 3\}$, *or* $(p, d) = (1951, 2)$. *Let* $j_{p,d} \in K_d$ *be Klein's invariant, computed as in §5, of the element* $y = u + \sqrt{-d}\,v$ *of Corollary 8.2 with the* u, v *given in the supplement, and let* E/K_d *be an elliptic curve with* $j(E) = j_{p,d}$. *For* $p = 1951$ *and* $d \in \{1, 3\}$ *these are the curves of Theorem C.*

(i) *The field of definition of a* 5*-isogeny of* E *is* K_dK_6. *So* $E[5] \otimes \bar{\mathbb{F}}_5 \cong \bar{\rho}|_{G_{K_d}} \otimes \psi$ *for a character* ψ.
(ii) $v(j_{p,d}) < 0$ *exactly at the places in Table 2, and* $v(j_{p,d}) = -5$ *at those not above* p. *Elsewhere* E *has potentially good reduction.*
(iii) E *is modular.*

Proof. As for Theorem C. In each case the fibre polynomial is irreducible over K_d, its stem field has degree 24 and contains K_dK_6 (`nfisincl`), and the hypotheses of [11, §6.1] hold, with the decomposed-generic witnesses of Table 2. As a further check, $a_P(E)^2 \equiv N(P)\,\bar{t}_\ell^2 \pmod 5$ at every degree-one prime $P \nmid p$ of good reduction with $N(P) = \ell < 3000$. □

TABLE 2. The curves of Theorem E (for $p = 1951$ and $d = 1, 3$ see Theorem C): the primes $\ell \neq p$ with a place where $v(j) = -5$ (two such places where marked), the number of places above p with $v(j) < 0$, the smallest decomposed-generic witness ℓ, and the Frobenius check in the proof.

p	d	$\ell \neq p$ with $v(j) = -5$	above p	witness	Frobenius
1951	2	569, 18371, 30859, 7091231370161	2	19	403/403
2141	1	29 (2), 149, 4552789, 3862149761	4	41	385/385
2141	3	19 (2), 829, 1566199, 1721658659471131	2	31	388/388
3701	1	41, 5281, 292480791140893601	4	29	389/389
3701	3	151808280432187244497986065351	2	19	392/392
3821	1	541, 14741, 9220134010860181	4	29	390/390
3821	3	19, 3539317036217533699	2	79	390/390
8501	1	509, 1549, 954415467134858269	4	29	389/389
8501	3	19, 61, 4639, 4856192474278602729271	2	109	389/389
9461	1	89, 17056691749, 5332528102961	4	29	390/390
9461	3	31, 857807719, 3564692749	2	19	390/390

Theorem F. *Let $p \in \{2141, 3701, 3821\}$, let ε_p be the quadratic character of conductor p, which is even, and let S be $S_2(\Gamma_0(p), \chi)$ or $S_2(\Gamma_0(p), \chi\varepsilon_p)$.*

(i) *Statements (i)–(iii) of Theorem D hold for ρ and S, with the eigenvalues x, y of $\bar\rho_2(\mathrm{Frob}_2)$ and the dimensions of Table 3. In particular the joint eigenspace of T_3, T_5, T_7 is two-dimensional, and the generalised eigenspaces have dimensions 3 and 3. So in each of the two spaces exactly three newforms with $a_2 \equiv x$ and three with $a_2 \equiv y$ are congruent to ρ modulo the maximal ideal of $\bar{\mathbb{Z}}_2$.*

(ii) *There is a Katz cusp form $g_1^{(p)}$ of weight 1 on $\Gamma_1(p)$ over $\mathbb{F}_{16}$, on which the diamond operators act through $\bar\chi$, with $a_1 = 1$ and $T_\ell g_1^{(p)} = \bar a_\ell g_1^{(p)}$ for every prime $\ell \nmid 2p$. No eigenform of weight one in characteristic 0 is congruent to it, as in Theorem D(v).*

Statement (iv) of Theorem D holds for every $p \in \mathcal{P}$.

Proof. The computations are those of Theorem D, in both spaces. The Sturm bounds are $(p+1)/6$ in weight 2 and $(p+1)/3$ in weight 4, and the coefficients were compared up to N_B of Table 3, which exceeds $p+1$. After clearing denominators in the PARI bases, saturation takes the number of rounds shown, and in each space the predicted forms for the other labelling of the classes of order 5, and for shuffled traces, do not occur. This fixes ρ_0 against ρ_0'', as in §2.

For (i) at all ℓ: by [39, 41], $\bar\rho_2$ comes from a newform h of weight 2 on $\Gamma_1(p)$ whose character reduces to $\bar\chi$. That character is $\chi\eta$ with η of 2-power order and conductor dividing p. Since $p \equiv 5 \pmod 8$, η is 1, ε_p, or one of two odd characters of order 4, so h lies in one of the two spaces. There the argument of Theorem D(i) applies. It transfers to the other space as follows.

- The forms G_x of the two spaces have the same q-expansion up to N_B. Both are Katz forms of weight 2 on $\Gamma_1(p)$ on which the diamond operators act through $\bar\chi$, since $\bar\varepsilon_p = 1$.
- Such a form whose q-expansion at ∞ vanishes to order greater than $(p+1)/6$ is zero. Indeed, it then vanishes to the same order at each of the $(p-1)/2$ cusps above ∞, which the diamond operators permute, and $(p-1)/2 \cdot (p+1)/6$ is the degree $(p^2-1)/12$ of ω^2 on $X_1(p)$.

So the two forms G_x coincide, and likewise the two forms G_y. Statement (ii) follows as in the proof of Theorem D(v), and (iv) has the same proof as there. □

TABLE 3. Theorem F, with Theorem D in the first row: the space, its dimension, the saturation rounds, the order of Frob_2 in A_5, the U_2-eigenvalues $x, y \in \mathbb{F}_{16} = \mathbb{F}_2(z)$, the dimensions of the joint eigenspaces of T_3, of $\{T_3, T_5\}$ and of $\{T_3, T_5, T_7\}$ (each becomes 1 after adding $T_2 - x$ or $T_2 - y$), the generalised eigenspaces for x and y, and the coefficient bound N_B.

p	S	dim	rounds	Frob_2	x, y	joint	generalised	N_B
1951	χ	161	4	3	$z+z^2$, $1+z^3$	4, 3, 2	3, 6	2400
2141	χ	177	5	5	z^3, z^4	4, 2, 2	3, 3	2600
2141	$\chi\varepsilon_p$	178	4	5	z^3, z^4	5, 3, 2	3, 3	2600
3701	χ	307	4	5	z^3, z^4	2, 2, 2	3, 3	4400
3701	$\chi\varepsilon_p$	308	4	5	z^3, z^4	3, 2, 2	3, 3	4400
3821	χ	317	4	5	z^2, z^3	2, 2, 2	3, 3	4600
3821	$\chi\varepsilon_p$	318	4	5	z^2, z^3	2, 2, 2	3, 3	4600

Remark 8.3. (1) For $p = 1951$ the two maximal ideals have ranks 3 and 6; in the six spaces of Theorem F both ranks are 3.

(2) For $p = 8501$ and 9461 the spaces $S_2(\Gamma_0(p), \chi)$ have dimensions 707 and 787. Constructing the space for 8501 in PARI exceeded the 8 GB of memory available to us (the run was stopped

by the system), and we did not attempt the larger space for 9461; so Theorem F is not stated for them. Lemma 8.1, Theorem E and Remark 8.4 cover all six representations.

Remark 8.4 (numerical: Stark's conjecture in rank 3). Let V and V' be the two irreducible three-dimensional representations of $\mathrm{Gal}(M/\mathbb{Q}) \cong A_5$, namely $\mathrm{Ad}\,\rho_0$ and $\mathrm{Ad}\,\rho_0''$, and let σ be the non-trivial automorphism of $\mathbb{Q}(\sqrt{5})$. As M is totally real, $L(s,V)$ has a zero of order $r(V) = \dim V = 3$ at $s = 0$. Let $c(V)$ be the leading coefficient there, for the full Artin L-function, whose Euler factor at p is $(1-p^{-s})^{-1}$ because $p \equiv 1 \pmod 5$. For a $\mathbb{Q}[G]$-isomorphism f from the degree-zero part of $\mathbb{Q}[\text{places of } M]$ to $\mathcal{O}_M^\times \otimes \mathbb{Q}$, let $R(V,f)$ be Stark's regulator. Stark's conjecture [58, Ch. I, §5] says that $A(V,f) = R(V,f)/c(V)$ lies in $\mathbb{Q}(\sqrt{5})$ and that $A(V',f) = \sigma(A(V,f))$.

We computed $R(V,f)$ through the fixed field K_{12} of a subgroup of order 5, on whose 12 real places V and V' each occur once. The Hecke algebra of K_{12} acts on the $(V \oplus V')$-parts of $\mathbb{Q}[\text{places of } K_{12}]$ and of $\mathcal{O}_{K_{12}}^\times \otimes \mathbb{Q}$ through $\mathbb{Q}(\sqrt{5})$. So sending three places to three units defines f on these parts, and $R(V,f)$ is a ratio of two 3×3 determinants. The check does not depend on the units being fundamental. The values $c(V)$ and $c(V')$ come from PARI's `lfun` at 57 digits, with Frobenius classes computed as in §2 and the functional equation checked by `lfuncheckfeq`. Table 4 summarises the results. For every $p \in \mathcal{P}$ and every non-degenerate choice of f, $A(V,f)$ agrees with an element $a + b\sqrt{5}$ of $\mathbb{Z}[\varphi]$, and $A(V',f)$ with $a - b\sqrt{5}$, to within 10^{-43}. Typical values are 4 and $\pm 2 \pm 2\sqrt{5}$.

Tate proved Stark's conjecture for rational-valued characters [58, Ch. II], so the rationality of $A(V,f)A(V',f)$ is known. The content checked here is that $A(V,f)$ lies in $\mathbb{Q}(\sqrt{5})$ and is σ-equivariant. We have not found earlier numerical checks of Stark's conjecture for a non-abelian character of rank greater than 1. Jehanne, Roblot and Sands [34] treat odd icosahedral representations, in rank 1, and McGown, Sands and Vallières [46] treat abelian characters of higher rank.

TABLE 4. Remark 8.4: the class number of K_{12}, the number of Dirichlet coefficients, the bit accuracy of the functional equations of $L(s,V)$ and $L(s,V')$, the number of non-degenerate choices of f, the largest $\log_{10}$ distance of $(A(V,f) + A(V',f))/2$ and $(A(V,f) - A(V',f))/2\sqrt{5}$ from the integers a, b, and the values of $|N(A(V,f))|$.

p	$h(K_{12})$	coefficients	`lfuncheckfeq`	choices	$\log_{10}$ residual	$\lvert N(A)\rvert$
1951	1	240000	-187, -188	28	-47.2	16, 64
2141	1	450000	-186, -188	15	-47.0	16, 64
3701	2	240000	-184, -183	26	-47.3	4, 16, 20
3821	4	240000	-186, -185	44	-45.9	4, 16, 20
8501	1	500000	-187, -185	27	-43.6	16, 64
9461	1	550000	-185, -185	29	-43.2	16, 64

9. QUESTIONS

1. Are Theorems A, A′ and A″ recorded in the literature? Theorem A″ fixes the sign of Lemma 3.4 with one cyclic cubic field, and Proposition 3.9 shows that finitely many fields without real places cannot fix it through local components. Is there a global argument for the sign? In particular, does the automorphy of $\mathrm{Ad}\,\rho$, which follows from H1 for one quadratic field, imply the modularity of ρ, as stated in [4, §2]?

   What we found. We have not found Theorems A, A′ or A″ in the literature, and we still do not see how the result of Flicker cited in [4, §2] gives the step from $\mathrm{Ad}\,\rho$ to ρ: the SL(2)

packet does not see the sign (Remark 3.6(3)). Booker's later paper [5, §1] says that obtaining an Artin representation from a Maass form of icosahedral type would need the expected cuspidality criterion for the symmetric sixth power. In the terms of Lemma 3.3: if H1 holds for one quadratic field E, with descent $\Pi_{\mathbb{Q}}$, then ρ is modular if and only if $\Pi_{\mathbb{Q}}$ corresponds to some Artin representation ρ'. Indeed, then $\rho'|_{G_E} \cong \rho|_{G_E}$, so ρ' is ρ or $\rho \otimes \varepsilon_E$.

2. Is any case of base change for GL(2) along a non-normal quintic extension with Galois closure A_5 known for π not already known to be attached to a Galois representation? Is it known conditionally only on automorphy of the four-dimensional representation of A_5, or of automorphic induction from L_5?

   What we found. We know of no such case. Conditionally, automorphic induction from L_5 to GL(5) for all Hecke characters λ of L_5 gives H2 through the converse theorem over L_5, since $L(s, \mathrm{BC}(\pi) \otimes \lambda) = L(s, \pi \times \mathrm{AI}(\lambda))$. Automorphy of **4** alone gives only the twists by $\lambda = \mu \circ N_{L_5/\mathbb{Q}}$, for which $\mathrm{AI}(\lambda) = \mu \boxplus \mu\mathbf{4}$, and these do not suffice (Remark 3.12(3)).

3. *The missing step in Theorem A′.* Let $\mathcal{F}_7$ be the set of fields $\mathbb{Q}(\sqrt{-q})$ with $q \equiv 7 \bmod 8$ prime. The prime 2 splits in each of them, and every odd prime p splits in one of them, since by Dirichlet's theorem there is such a q with $\left(\frac{-q}{p}\right) = 1$; for $p < 10^6$ the first fifteen fields suffice. So $\mathcal{F}_7$ satisfies the hypothesis of Theorem A′. For each $E \in \mathcal{F}_7$ the following hold.
   - $\bar\rho_2|_{G_E}$ is automorphic in cohomological weight. It is the reduction of the base change to E of the newforms of Theorem D(iii), which remains cuspidal because their projective image is A_5.
   - At $v \mid 2$ we have $E_v = \mathbb{Q}_2$, $\rho|_{G_{E_v}}$ is unramified, and $\bar\rho_2(\mathrm{Frob}_v)$ has the distinct eigenvalues x and y of §6.
   - At 1951, inertia acts on ρ and on $\bar\rho_2$ through the same group of order 5, so ρ is minimally ramified.

   Is there a modularity lifting theorem that applies to $\rho|_{G_E}$ under these hypotheses, that is, to a 2-adic representation of G_E all of whose Hodge–Tate weights are 0? Applied to every $E \in \mathcal{F}_7$, it would prove that ρ is modular, by Theorem A′.
   - A representation corresponding to $\rho|_{G_E}$ would have archimedean parameter of Artin type, which contributes to no cohomology of Bianchi groups.
   - Over $\mathbb{Q}$, $\bar\rho_2$ comes from the weight-one form g_1, which has no lift of weight one (Theorem D(v) and Remark 6.2(3)).
   - Lifting theorems over CM fields are usually stated for $p > 2$ and cohomological weight. The method of [10] reaches weight one over $\mathbb{Q}$ for odd p through coherent cohomology, which has no analogue for GL_2 over E. The device for even representations described in [7, §9.7] needs distinct Hodge–Tate weights. Theorem B rules out a mod-5 elliptic starting point over quadratic fields.
   - The numerical defect is 3. For Hodge–Tate weights 0 the local terms at p in the Greenberg–Wiles formula vanish, so the defect of the patching method is

     $$\ell_0 = \textstyle\sum_{v|\infty} h^0(G_{E_v}, \mathrm{ad}^0\rho) = 3,$$

     the rank of the $\mathrm{ad}^0\rho$-part of the units of ME. It is also 3 for ρ over $\mathbb{Q}$, since $\rho(c) = -I$ acts trivially on $\mathrm{ad}^0\rho$, and it is $3(r_1 + r_2)$ over a field with r_1 real and r_2 complex places. Odd weight one over $\mathbb{Q}$ and cohomological weight over E have $\ell_0 = 1$. So passing to E removes the parity obstruction but not the defect.
   - First-order deformations of $\rho|_{G_E}$ are anti-invariant under $\mathrm{Gal}(E/\mathbb{Q})$, for $p \in \{2, 3, 5, 7\}$. Let K_{12} be the fixed field of a subgroup of order 5 of $\mathrm{Gal}(M/\mathbb{Q})$, defined by $x^{12} - 113x^{10} + 4482x^8 - 72251x^6 + 388968x^4 - 48712x^2 + 1225$, of discriminant 1951^8. The p-adic logarithms of its units have rank 11 for these p, so Leopoldt's conjecture holds for K_{12} at p. The $\mathrm{ad}^0\rho$-part of the units of M has non-zero invariants under that subgroup, so it

injects p-adically, and $H^1(G_{\mathbb{Q},S}, \mathrm{ad}^0\rho \otimes \bar{\mathbb{Q}}_p) = 0$ for every finite S. Hence $\mathrm{Gal}(E/\mathbb{Q})$ acts by -1 on $H^1(G_{E,S}, \mathrm{ad}^0\rho \otimes \bar{\mathbb{Q}}_p)$. For nearly ordinary deformations whose ordinary lines at the two places above a split prime correspond under $\mathrm{Gal}(E/\mathbb{Q})$, as they do for base change, the infinitesimal weight is therefore anti-parallel, while cohomological Bianchi forms have parallel weights. This is the even case of [9, Cor. 1.4], there conditional on a strong Leopoldt conjecture; by [8], under its hypotheses, G_E has no irreducible ordinary two-dimensional representations of non-parallel weight. With the other choice of ordinary lines the first-order picture is different (Proposition 9.1).

What we found. We know of no such theorem. Four remarks on the obstruction:

- Galois representations attached to torsion classes in the cohomology of Bianchi manifolds exist [54], and local–global compatibility is known in many cases [1]. This part of the programme of [10] is not what is missing for $\rho|_{G_E}$.
- The size of the defect is not by itself the obstruction. For a totally odd two-dimensional Artin representation of G_F, with F totally real of degree d, the defect is $\ell_0 = d$, one for each real place. Modularity lifting theorems in parallel weight one are nevertheless known [36, 37, 48, 53]. They work on Hilbert modular varieties, through Hida families and the analytic continuation of overconvergent forms. For GL_2 over E there is no Shimura variety, and forms of Artin type contribute to no cohomology, so neither route has a place to start.
- The defect is the order of vanishing at $s = 0$: $r(\mathrm{ad}^0\rho) = \sum_{v|\infty} h^0(G_{\mathbb{Q}_v}, \mathrm{ad}^0\rho) = 3 = \ell_0$. By Stark's conjecture, the leading term of $L(s, \mathrm{ad}^0\rho)$ there is the regulator of the $\mathrm{ad}^0\rho$-part of the units of M, up to a factor in $\mathbb{Q}(\sqrt{5})$; Remark 8.4 checks this numerically for all six representations of §8. For odd weight one this part has rank $\ell_0 = 1$, and Harris and Venkatesh [29] conjecture that its Stark unit governs the action of the derived Hecke algebra [60]. Here the corresponding lattice has rank 3.
- For all six representations of §8, Leopoldt's conjecture holds for their fields K_{12} at $\ell = 2, 3, 5, 7$: the ℓ-adic logarithms of the units have rank 11 in all 24 cases. So $H^1(G_{\mathbb{Q},S}, \mathrm{ad}^0\rho \otimes \bar{\mathbb{Q}}_\ell) = 0$, and the anti-invariance above holds for them. What it implies for ordinary families depends on the choice of ordinary lines.

Proposition 9.1 (nearly ordinary tangent spaces)**.** *Let ρ be one of the six representations of §8, of conductor p, and let E be an imaginary quadratic field. Let $\ell \neq p$ be a prime that splits in E, say $\ell\mathcal{O}_E = v\bar{v}$, such that $\rho(\mathrm{Frob}_\ell)$ is not scalar, with eigenvalues $\alpha \neq \beta$ and eigenlines L_α, L_β for Frob_v, and such that $H^1(G_{\mathbb{Q},S}, \mathrm{ad}^0\rho \otimes \bar{\mathbb{Q}}_\ell) = 0$ (for instance $\ell \leq 7$, or $\ell \in \{11, 31, 41, 61, 71\}$). Fix $\sigma \in G_\mathbb{Q}$, non-trivial on E, with $\sigma D_v \sigma^{-1} = D_{\bar{v}}$. Consider first-order deformations of $\rho|_{G_E}$ with fixed determinant that are nearly ordinary for a line L_v at v and a line $L_{\bar{v}}$ at $\bar{v}$. Their infinitesimal weight is the pair of derivatives of the characters by which inertia acts on these lines.*

(i) *$H^1(G_{E,S}, \mathrm{ad}^0\rho \otimes \bar{\mathbb{Q}}_\ell)$ has dimension 3, and restriction maps it isomorphically onto $H^1(D_v, \mathrm{ad}^0\rho)/H^1_{\mathrm{ur}}(D_v, \mathrm{ad}^0\rho)$.*

(ii) *If $L_v = L_\alpha$ and $L_{\bar{v}} = \rho(\sigma)L_\alpha$, as for base change, the nearly ordinary tangent space has dimension 2, and its infinitesimal weights are $(w, -w)$.*

(iii) *If $L_v = L_\alpha$ and $L_{\bar{v}} = \rho(\sigma)L_\beta$, it has dimension 1, and its infinitesimal weights are (w, w) with $w \neq 0$.*

Proof. By Shapiro's lemma and the hypothesis, $H^1(G_{E,S}, \mathrm{ad}^0\rho) \cong H^1(G_{\mathbb{Q},S}, W)$ with $W = \mathrm{ad}^0\rho \otimes \varepsilon_E$. Every class c satisfies $c(\sigma h \sigma^{-1}) = -\mathrm{Ad}\rho(\sigma)\, c(h)$ for $h \in D_v$, up to coboundaries. Let $N = ME$ and $G = \mathrm{Gal}(N/\mathbb{Q})$.

- Complex conjugation acts on W by -1, so by Herbrand's theorem W does not occur in $\mathcal{O}_N^\times \otimes \bar{\mathbb{Q}}_\ell$.

- Classes ramified at a prime $q \neq \ell$ do not occur, because the eigenvalues of Frobenius on W^{I_q} are roots of unity.
- So class field theory gives $H^1(G_{\mathbb{Q},S}, W) \cong \mathrm{Hom}_G(U, W)$. Here U is the group of local units of N at ℓ, completed and tensored with $\bar{\mathbb{Q}}_\ell$, a free $\bar{\mathbb{Q}}_\ell[G]$-module of rank one since ℓ is unramified in N.
- By Frobenius reciprocity, restriction to the inertia subgroup of D_v is an isomorphism, which gives (i).

The nearly ordinary condition at a place is the vanishing of the component in $H^1(D, \psi)$, where $\psi = \mathrm{Hom}(L, V/L)$. This space is one-dimensional and consists of ramified classes, since ψ is unramified and non-trivial. In case (ii) the conditions at v and $\bar{v}$ coincide, by the formula for $c(\sigma h \sigma^{-1})$. In case (iii) they are the vanishing of both off-diagonal components at v. The weight at v is the diagonal entry of $c|_{D_v}$ on L_α, restricted to inertia, and the same formula gives the weight at $\bar{v}$. It is minus the entry on L_α in case (ii), and minus the entry on L_β, which is plus the entry on L_α, in case (iii). In case (iii) the remaining class is diagonal and, by (i), non-zero on inertia, so $w \neq 0$. □

So the anti-parallel conclusion in the item on first-order deformations above concerns the base-change choice of lines, case (ii); the version of 19 October stated it without that restriction. In case (iii), $\rho|_{G_E}$ has a one-dimensional nearly ordinary tangent space that maps isomorphically to the parallel direction of weight space. That is what a Hida family of parallel weight through $\rho|_{G_E}$ would need at first order. Its classical members would be Bianchi forms that are not base changes, since U_v and $U_{\bar{v}}$ differ residually; this is the exceptional "even" case allowed for in [9, §1].

- At an auxiliary prime $\ell \equiv 1 \pmod 5$ that splits in E, with $\rho(\mathrm{Frob}_\ell)$ not scalar, Proposition 9.1 applies as soon as Leopoldt's conjecture holds for K_{12} at ℓ. For the fields K_{12} of all six representations it holds at $\ell = 11, 31, 41, 61, 71$: the ℓ-adic logarithms of the units have rank 11 in all 30 cases. For instance, 11 and 71 split in $\mathbb{Q}(\sqrt{-7})$, and for each of the six representations Frob_{11} and Frob_{71} are not central. But the residual representation is even over $\mathbb{Q}$, and it is not known to be modular over E. The residual input over E is available at $\ell = 2$, and at $\ell = 5$ only over the fields K_d of Theorem E.
- At $\ell = 2$ and $E \in \mathcal{F}_7$, the residual input for case (iii) would be a mod-2 Bianchi eigenclass of level $1951\mathcal{O}_E$ in the system of $\bar{\rho}_2|_{G_E}$, with eigenvalues x and y for the Hecke operators at v and $\bar{v}$. Base change of the forms of Theorem D gives only (x, x) and (y, y). For $E = \mathbb{Q}(\sqrt{-7})$ we computed $H^1(\Gamma_0(1951\mathcal{O}_E), \mathbb{F}_{16})$ for $\mathrm{PGL}_2(\mathcal{O}_E)$ with the nebentypus $\bar{\chi}^{-1} \circ N$, using the presentation of $\mathrm{PSL}_2(\mathcal{O}_E)$ in [20, (2.10)] and its Hecke operators. The index of $\Gamma_0(1951\mathcal{O}_E)$ is $1951^2 + 1 = 3\,806\,402$. The linear algebra is a sparse elimination followed by a block Lanczos solver over $\mathbb{F}_{16}$. Every cocycle found satisfies all $3.0 \cdot 10^7$ relator equations, a Krylov-dimension bound certifies that none is missing, and a second run of the solver with other random choices gives the same space. The space has dimension 663. Its generalised eigenspace for $T_{(3)}, T_{(5)}, T_{(13)}, T_{(\sqrt{-7})}, T_{\mathfrak{q}}, T_{\bar{\mathfrak{q}}}$ ($N\mathfrak{q} = 11$) at the eigenvalues of $\bar{\rho}_2$ has dimension 27, with multiplicities $7, 3, 3, 14$ for $(T_v, T_{\bar{v}}) = (x, x), (x, y), (y, x), (y, y)$. Each of these four generalised eigenspaces contains exactly one eigenvector with the eigenvalues of $\bar{\rho}_2$ at all 24 Hecke operators computed. These are T_v, $T_{\bar{v}}$, $T_{(3)}$, $T_{(5)}$, $T_{(13)}$, $T_{(\sqrt{-7})}$, and $T_{\mathfrak{q}}$, $T_{\bar{\mathfrak{q}}}$ for the split primes $q \leq 79$. The four vectors are also eigenvectors, with the eigenvalues of $\bar{\rho}_2$, at 26 further operators: $T_{\mathfrak{q}}$ and $T_{\bar{\mathfrak{q}}}$ for the split primes $79 < q \leq 200$, and $T_{(17)}$, $T_{(19)}$. So they have the eigenvalues of $\bar{\rho}_2$ at every prime of E of norm at most 200. So for $E = \mathbb{Q}(\sqrt{-7})$ classes with $(T_v, T_{\bar{v}}) = (x, y)$ and (y, x) exist at this level, with the eigenvalues of $\bar{\rho}_2$ at every operator tested. That they belong to the system of $\bar{\rho}_2|_{G_E}$ for all Hecke operators is not proved. They do not lift to characteristic 0 with the nebentypus of the computation. Let χ be the Teichmüller

lift of $\bar{\chi}$. Over $\mathbb{F}_P$, $P = 1\,048\,361$, a sparse elimination followed by a block Lanczos solver over $\mathbb{F}_P$ bounds $\dim H^1(\Gamma_0(1951\mathcal{O}_E), \mathbb{F}_P(\chi^{-1} \circ N))$ by 161 (Krylov-dimension bound). This bounds the dimension in characteristic 0. Base change of the 161 newforms of $S_2(\Gamma_0(1951), \chi)$ gives 161 classes. So in characteristic 0 this cohomology consists of base changes, and $T_v = T_{\bar{v}}$ on it. The mixed classes therefore come from 2-power torsion. They lift modulo 4, so this torsion has order at least 4. The module $\mathbb{F}_{16}(\bar{\chi}^{-1} \circ N)$ is also the reduction of the modules with nebentypus $(\chi^{-1} \circ N)\psi$, for every character ψ of $(\mathcal{O}_E/1951)^\times$ of 2-power order with $\psi(-1) = 1$. For the quadratic twist $\psi = (\frac{\cdot}{1951}) \circ N$, base change from $S_2(\Gamma_0(7 \cdot 1951), \chi(\frac{\cdot}{1951})\chi_{-7})$ gives 649 classes; we have not yet compared this with the dimension over $\mathbb{F}_P$. For ψ of order 4 to 32 we have no computation in characteristic 0. The first-order 2-adic obstruction of each of the four eigenvectors (lifting modulo 4, or modulo ϖ^2 when ψ has order at least 4) vanishes for every ψ, so that test does not decide these cases. We do not know whether the deformation theory of case (iii) can be run from these classes at $\ell = 2$. The same pattern occurs in smaller models, where both refinements occur over $\mathbb{Q}$. For the dihedral mod-2 representations of the weight-2 newforms of levels $N = 23, 31, 47, 79, 87, 93, 103, 115, 127, 143$, the $\bar{\rho}$-part at level $N\mathcal{O}_E$ contains eigenclasses with $(T_v, T_{\bar{v}}) = (x, y)$ and (y, x), besides (x, x) and (y, y). For $N = 23, 31, 47$ the mixed classes are reductions of no eigenform of characteristic 0 at that level, for any nebentypus of 2-power order. With trivial nebentypus the characteristic-0 cohomology consists of base changes. For the quadratic twists the obstruction to lifting the mixed classes modulo 4 is non-zero. For the twists of order at least 4 (orders 4 to 32 at 31 and 4 to 16 at 47; there are none at 23) we computed the characteristic-0 cohomology, and $T_v + T_{\bar{v}}$ is nilpotent on it modulo the prime above 2. Searches over $\mathbb{Q}$ found no other icosahedral model. They cover odd squarefree $N \leq 400$ with trivial character, and odd $N \leq 150$ with every nebentypus whose odd part has order dividing 15; nine newforms at levels above 95, with coefficient fields of degree at least 80, were too large for us to reduce. The models found are dihedral or reducible, and icosahedral systems at these levels have no companion. The weight-one newform of level 633 with projective image A_5, the smallest such level [6], reduces to a model of the same type as $\bar{\rho}_2$: its nebentypus has order 5 modulo 2, and $\bar{\rho}(\mathrm{Frob}_2)$ has projective order 5. At level $633\mathcal{O}_E$ (index 449 440) it gives the same picture, with one joint eigenvector at 23 operators in each of the four pairs. With the Teichmüller lift of its nebentypus, the characteristic-0 cohomology at $633\mathcal{O}_E$ consists of the 68 base changes (bounds over $\mathbb{F}_P$ for two primes), so there the mixed classes do not lift with that nebentypus. An icosahedral system that is ramified at 2 (level 31, nebentypus of order 5) has only $(T_v, T_{\bar{v}}) = (a_2, a_2)$ at level $31\mathcal{O}_E$.

4. Have curves such as those in Theorem C, built from Klein's accessory irrationality over CM quartic fields, been recorded for other even icosahedral representations?

   What we found. We have not found them. By Lemma 8.1 and Corollary 8.2, Klein's construction starts over every imaginary quadratic field for every totally real quintic field of discriminant p^4 with $p > 5$. Theorem E completes and certifies it for the six representations of [17], with $d = 1, 3$, and with $d = 2$ for $p = 1951$.

5. Has the weight-one form g_1 of Theorem D(v), or its analogue for other even icosahedral representations, been recorded?

   What we found. We have not found g_1 or an analogue recorded; the weight-one forms modulo 2 known to us are those of Remark 6.2(2). Theorem F gives the analogues for $p = 2141$, 3701 and 3821. For 8501 the computation exceeded our memory, and 9461 was not attempted (Remark 8.3).

Data and code

The supplement contains the scripts and outputs behind every computational statement, with a README that maps each statement to its script and output:

- the trace tables and their generator, and the check that K_{24}/K_6 is unramified at finite places;
- the exact j_0 and j_1;
- the Klein resolvent at 600 digits and the field-equality certificates;
- the Frobenius and witness checks;
- the saturation and Hecke-module computations modulo 2, and checks of the identities in $\mathbb{F}_{16}$ used in Theorem D(v);
- the characteristic-0 determinant check;
- the character computations behind Remark 3.11;
- the Hilbert-symbol and certifying-prime computations;
- the Hejhal computation and the Galois series over $\mathbb{Q}$ cited in the introduction (numerical);
- the restricted Weil form, its ball-arithmetic certificate and the controls of Remark 7.2;
- the special values and p-adic checks of Remark 7.4;
- the Bianchi series and the series over $\mathbb{Q}(\sqrt{2})$ of Remark 3.10, with the trace table extended to $4.4 \cdot 10^8$ (numerical), and the field K_{12} with the p-adic logarithms of its units used in Question 3 and Proposition 9.1;
- the cohomology of $\Gamma_0(N\mathcal{O}_E)$ in $\mathrm{PSL}_2(\mathcal{O}_E)$ and $\mathrm{PGL}_2(\mathcal{O}_E)$, $E = \mathbb{Q}(\sqrt{-7})$, modulo 2 and in characteristic 0, with Hecke operators and checks against LMFDB, also with coefficients in $\mathbb{F}_{16}$ and nebentypus at the levels $31\mathcal{O}_E$, $633\mathcal{O}_E$ and $1951\mathcal{O}_E$ (sparse elimination and block Lanczos over $\mathbb{F}_{16}$), and the searches for models over $\mathbb{Q}$, used in Question 3;
- for Section 8: the Frobenius classes, Klein invariants and certificates, the spaces modulo 2, the fields K_{12} and the Stark checks (numerical) for the six conductors, with a summary file.

All exact computations use PARI/GP [47], except the character computations, which use GAP [22], the ball arithmetic of Proposition 7.1, which uses Arb [35] through python-flint, and the linear algebra over $\mathbb{F}_2$, $\mathbb{F}_{16}$ and $\mathbb{F}_P$ for the Bianchi cohomology in Question 3, which uses our own code (numpy and numba).

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